Data_Analysis_Code

Analysis Code and Narrative Descriptions

Goals

Broad question

What social factors are important in female mate choice?

Specific question(s)

Do interactions with certain male reproductive tactics increase a female's propensity to spawn at a nest?

Does nesting male courtship behavior increase the propensity for a female to visit or spawn with that male?

How does nesting male aggression influence female visits and spawning?

This document

The purpose of this document is to catch you up o the progress of our research on mate choice! It will help you visualize what we've been doing and what we've found! Please ask questions, there are no wrong questions we are all learning together!

Tidying code/Wrangling

Here we are downloading and getting all the important libraries we will need for the code. Then we are going to download the raw data. Then we fix some of the columns so we can use them for analysis, especially the date column!

```r
#Install and library important packages
#install.packages('ggplot2')
#install.packages('ggeffects')
#install.packages('RColorBrewer')
#install.packages('here')
#install.packages('MASS')
#install.packages('tidyverse')
#install.packages('janitor')
#install.packages('readxl')
#install.packages('brms')
#install.packages('ProbBayes')
library(ProbBayes)
```

```
Loading required package: LearnBayes

Loading required package: ggplot2

Loading required package: gridExtra

Loading required package: shiny
```

```r
library(ggeffects)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v lubridate 1.9.4     v tibble    3.3.1
v purrr     1.2.1     v tidyr     1.3.2

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::combine() masks gridExtra::combine()
x dplyr::filter()  masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(here)
```

```
here() starts at /Users/jacobchung/coding_projects/Owrasse_Underwater_Behavioral_Sheet
```

```r
library(ggplot2)
library(readxl)
library(dplyr)
library(brms)
```

```
Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').

Attaching package: 'brms'

The following object is masked from 'package:LearnBayes':

    rdirichlet

The following object is masked from 'package:stats':

    ar
```

```r
library(RColorBrewer)
library(tibble)
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```r
behavior_sheet <- read_excel("UW_Nest_Behavioral_Sheets_Cleaned.xlsx")

behavior_sheet_date <- as.Date(behavior_sheet$Date)
behavior_sheet$Date <- behavior_sheet_date
behavior_sheet_sep <- behavior_sheet %>%
  mutate(Year = year(Date)) %>%
  mutate(Month = month(Date)) %>%
  mutate(Day = day(Date)) %>%
  relocate(Year, .after = Date) %>%
  relocate(Month, .after = Year) %>%
  relocate(Day, .after = Month)
head(behavior_sheet_sep)
```

```
# A tibble: 6 x 73
  Date        Year Month   Day `Nest Name` `Start Time`        `Obs Length`
  <date>     <dbl> <dbl> <int> <chr>       <dttm>              <chr>
1 2025-05-21  2025     5    21 4           1899-12-31 10:19:00 10
2 2025-05-22  2025     5    22 OSL SS      1899-12-31 11:00:00 10
3 2025-05-23  2025     5    23 8           1899-12-31 10:14:00 10
4 2025-05-25  2025     5    25 25          1899-12-31 10:16:00 10
5 2025-05-25  2025     5    25 FOSR ZZ     1899-12-31 10:54:00 10
6 2025-05-26  2025     5    26 OSR YY      1899-12-31 10:12:00 10
# i 66 more variables: `NM Mark` <chr>, `SAT Mark` <chr>, Observer <chr>,
#   `Observation type` <chr>, `Average SNK` <dbl>, `# NM in` <dbl>,
#   `# NM near` <dbl>, `# NM far` <dbl>, `# SAT Present` <dbl>,
#   `# Sat Absent` <dbl>, `Female visits by NM` <dbl>,
#   `Female visits by SAT passed  to NM` <dbl>, `Female visit by SAT` <dbl>,
#   `Female visit by SNK` <dbl>, `Female spawning with NM brought by NM` <dbl>,
#   `Female spawning with only SAT brought by NM` <dbl>, ...
```

```r
#Fix SAT submission to NM from chr to num
sat_sub <- as.numeric(behavior_sheet_sep$`SAT submission to NM`) #Creates a vector with the s
```

```
Warning: NAs introduced by coercion
```

```r
behavior_sheet_sep$`SAT submission to NM` <- sat_sub #put it back into the tibble
#Do the same for SNK Submission to SAT
snk_sub <- as.numeric(behavior_sheet_sep$`SNK submission to SAT`)
```

```
Warning: NAs introduced by coercion
```

```
behavior_sheet_sep$`SNK submission to SAT` <- snk_sub
#Same for NM aggression to SAT
NM_aggsat <- as.numeric(behavior_sheet_sep$`NM aggression to SAT`)
```

```
Warning: NAs introduced by coercion
```

```
behavior_sheet_sep$`NM aggression to SAT` <- NM_aggsat

behavior_sheet_cln <- behavior_sheet_sep |>
  #Uses the Janitor package to standardize the names
  clean_names()

#Turn the year variable into a character for ease of comparison later on
year_date <- as.character(behavior_sheet_cln$year) #Creates a vector with years as chr
behavior_sheet_cln$year <- year_date #creates a new variable for ease of graphing later on
```

Visualization

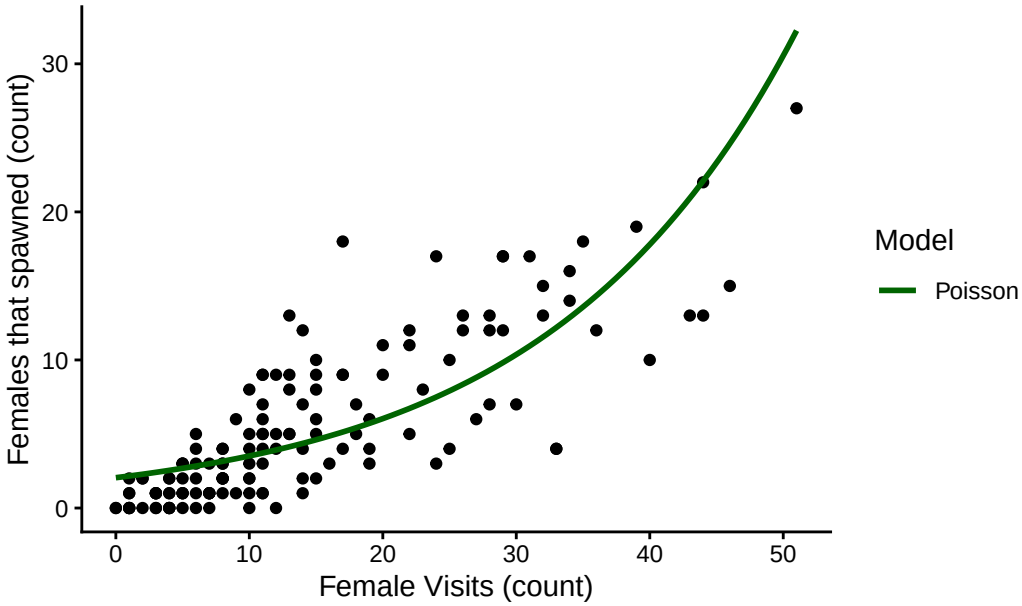
Female Spawns as a function of Female Visits

This is just to help us visualize what sort of processes are occurring at the nest.

```
fem.ts.fem.v <- behavior_sheet_cln |>
  ggplot(aes(total_female_visits, total_female_spawning)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth(method = 'glm', se = F, method.args = list(family = 'poisson'), aes(color = "Po
  scale_color_manual(name = 'Model', breaks = "Poisson", values = c('Poisson' = 'darkgreen')
  theme_classic() + #Sets the plot visuals
  labs(x = 'Female Visits (count)', y = 'Females that spawned (count)', title = "How many fem
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

fem.ts.fem.v
```

```
`geom_smooth()` using formula = 'y ~ x'
```

ow many females spawned as a function of total Female Visits

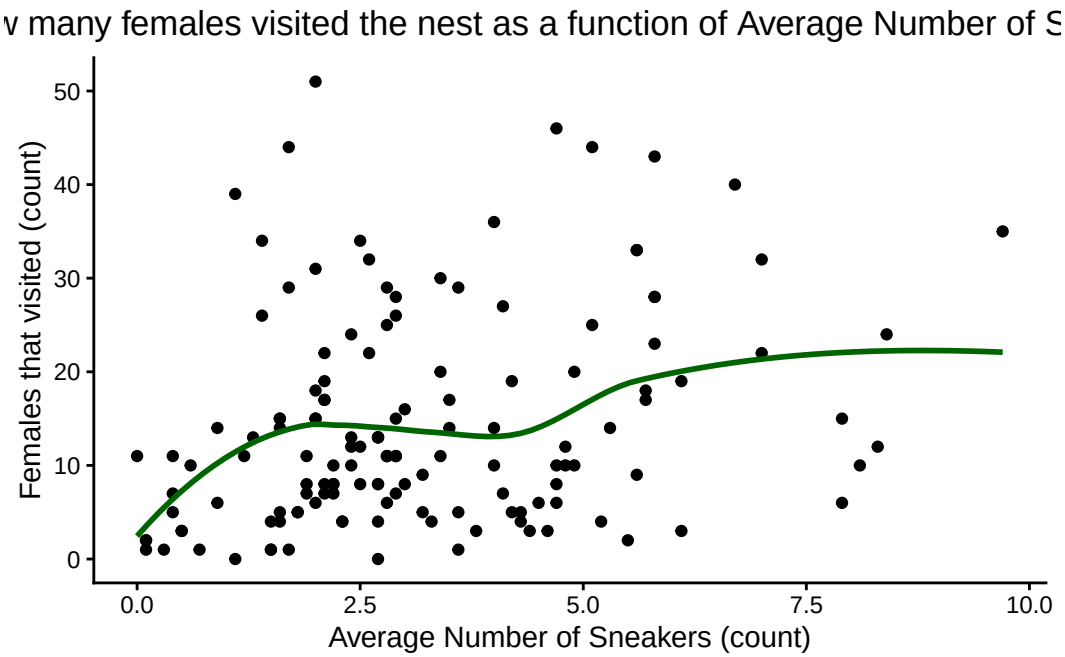


Female Visits/Spawns as a function of Average Number of Sneakers

Total Female visits as a function of Average Number of sneakers

```
fem.v.avg.snk <- behavior_sheet_cln |>
  ggplot(aes(average_snk, total_female_visits)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth(se = F, color = 'darkgreen') + #Adds a trendline for better visuals
  theme_classic() + #Sets the plot visuals
  labs(x = 'Average Number of Sneakers (count)', y = 'Females that visited (count)', title =
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

fem.v.avg.snk
```

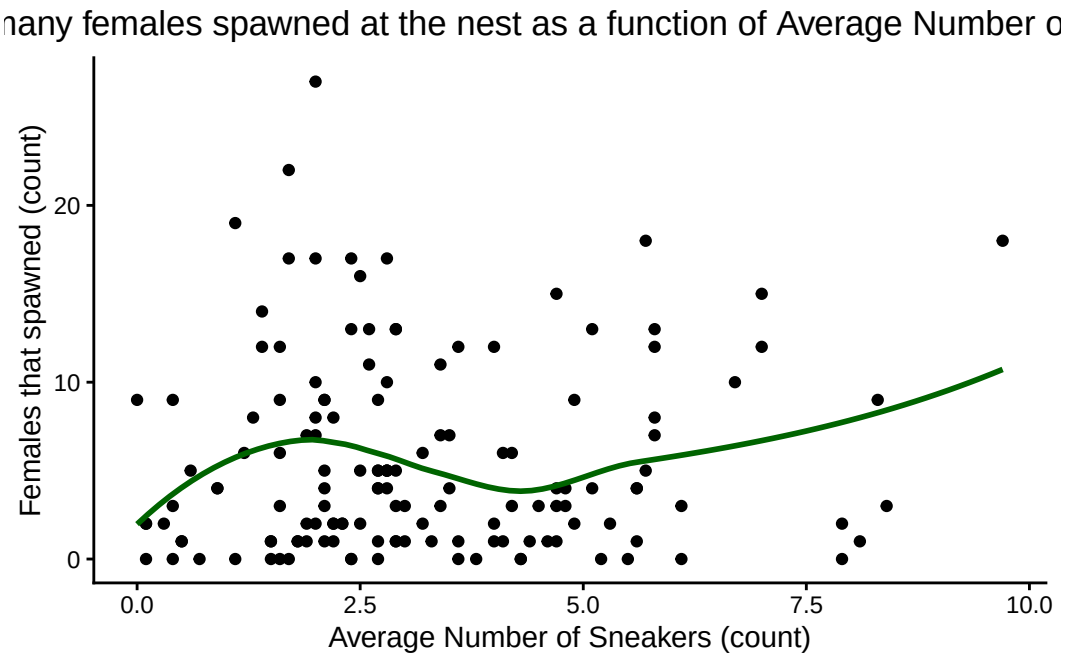
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Total Females spawned as a function of Average Number of sneakers

```
fem.s.avg.snk <- behavior_sheet_cln |>
  ggplot(aes(average_snk, total_female_spawning)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth(se = F, color = 'darkgreen') + #Adds a trendline for better visuals
  theme_classic() + #Sets the plot visuals
  labs(x = 'Average Number of Sneakers (count)', y = 'Females that spawned (count)', title =
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

fem.s.avg.snk
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

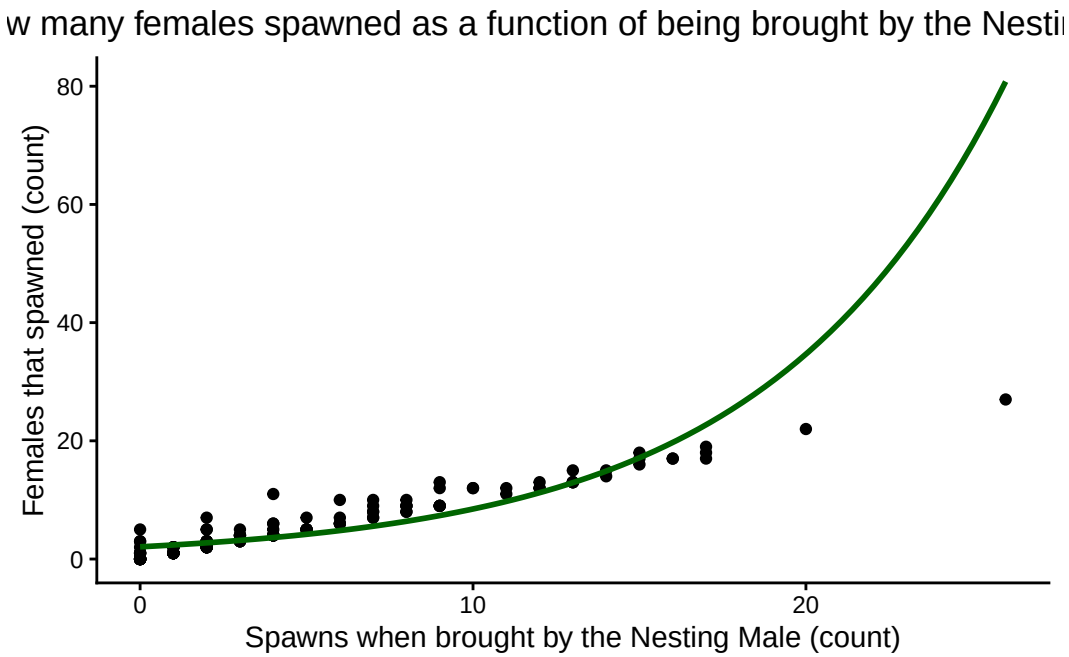


Female that spawns as a function of being brought to the nest by the Nesting Male

```
fem.s.bnm <- behavior_sheet_cln |>
  mutate('nm.brought.spawns'= (female_spawning_with_nm_brought_by_nm + female_spawning_with_
  ggplot(aes(nm.brought.spawns, total_female_spawning)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth(method = 'glm.nb', se = F, color = 'darkgreen') + #Plots a Negative Binomial mo
  theme_classic() + #Sets the plot visuals
  labs(x = 'Spawns when brought by the Nesting Male (count)', y = 'Females that spawned (cou
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

fem.s.bnm
```

```
`geom_smooth()` using formula = 'y ~ x'
```

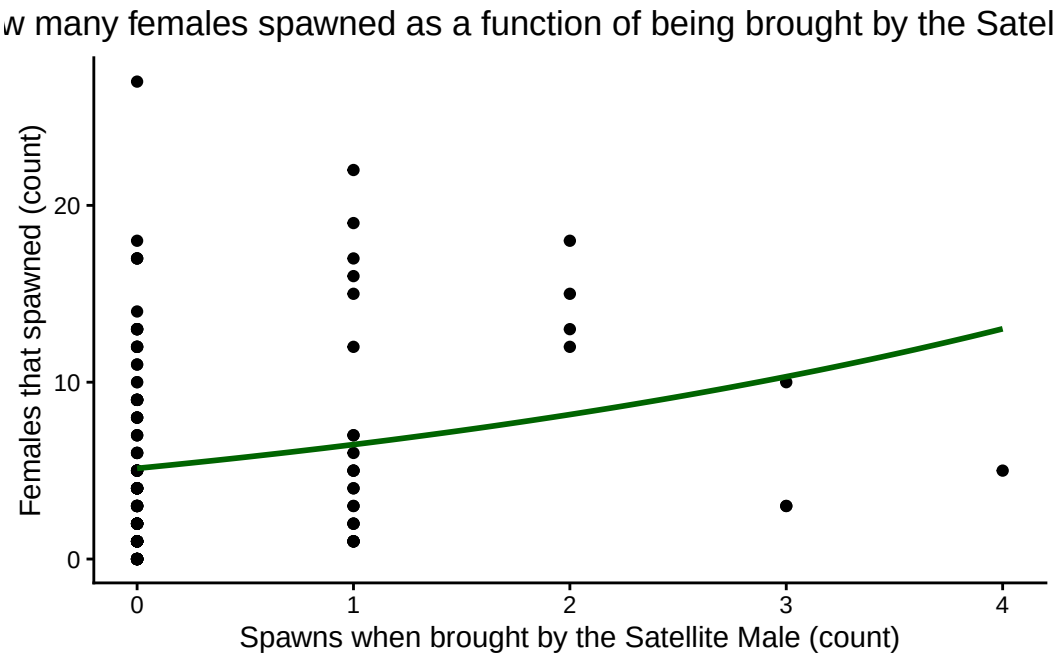
```
Warning in method(formula, data = data, weights = weight): alternation limit
reached
```



Female Spawns as a function of being brought to the nest by the Satellite Male

```
fem.s.bsat <- behavior_sheet_cln |>
  mutate('sat.brought.spawns'= (female_spawning_sat_brought + females_spawing_sat_brought_sat
  ggplot(aes(sat.brought.spawns, total_female_spawning)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth(method = 'glm.nb', se = F, color = 'darkgreen') + #Plots a Negative Binomial mo
  theme_classic() + #Sets the plot visuals
  labs(x = 'Spawns when brought by the Satellite Male (count)', y = 'Females that spawned (co
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

fem.s.bsat
```

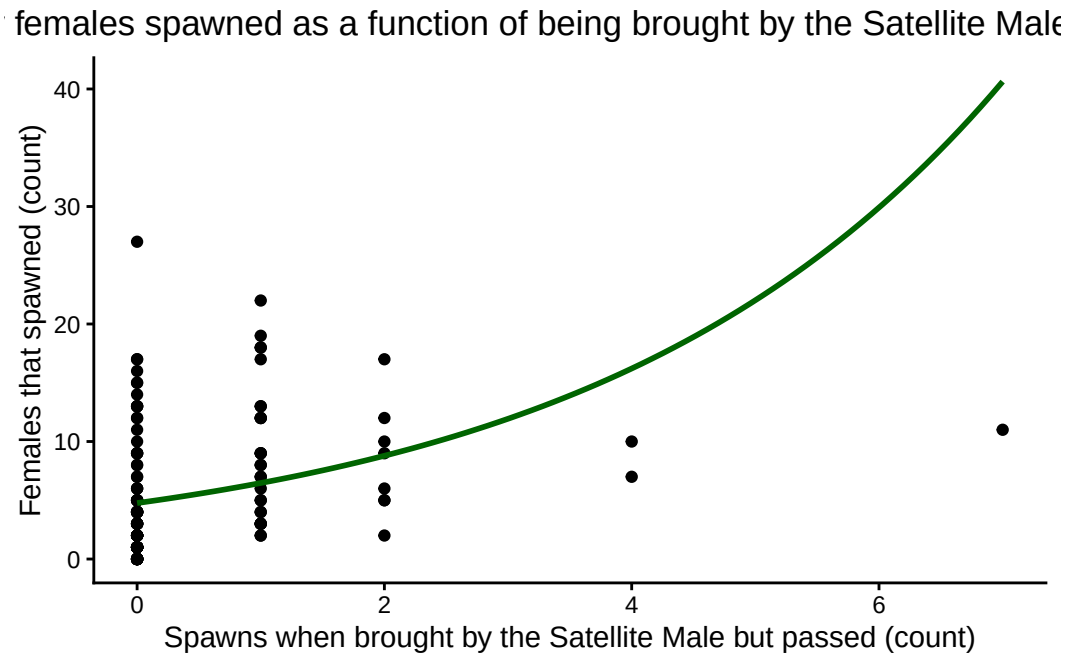
```
`geom_smooth()` using formula = 'y ~ x'
```



Female Spawns as a function of being brought to the nest by the Satellite Male and passed to the Nesting Male

```
fem.s.bsat.pnm <- behavior_sheet_cln |>
  mutate('bsat.pnm.spawns'= (female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to
  ggplot(aes(bsat.pnm.spawns, total_female_spawning)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
 geom_smooth(method = 'glm.nb', se = F, color = 'darkgreen') + #Plots a Negative Binomial mod
  theme_classic() + #Sets the plot visuals
  labs(x = 'Spawns when brought by the Satellite Male but passed (count)', y = 'Females that
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

fem.s.bsat.pnm
```

```
`geom_smooth()` using formula = 'y ~ x'
```

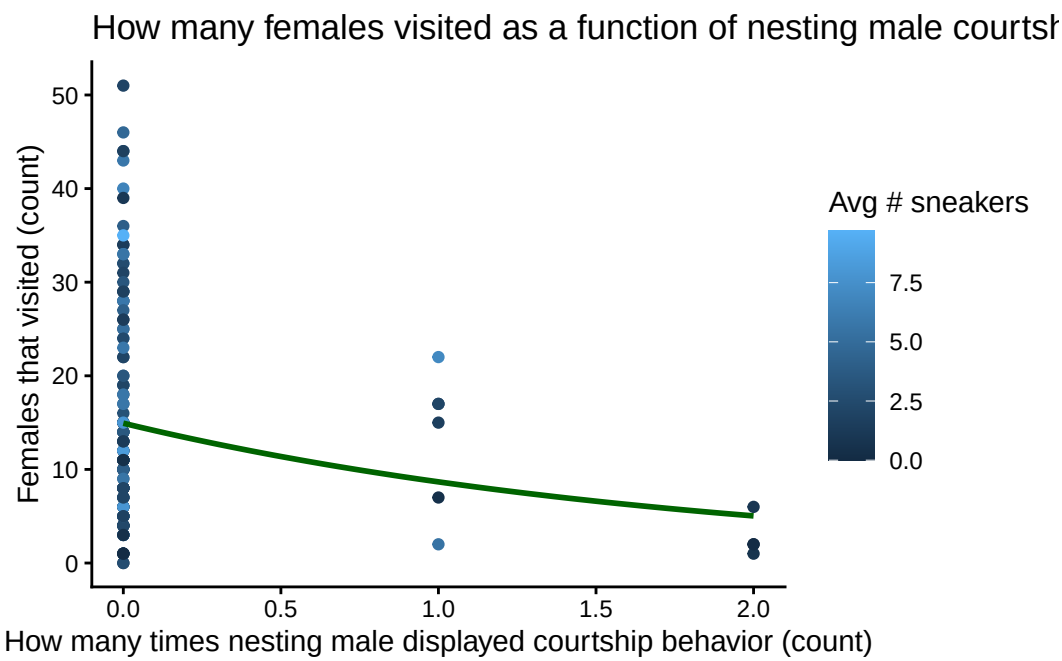


Female Visits/Spawns as a function of Nesting Male Courtship

Total Female Visits as a function of Nesting Male Courtship

```
fem.v.nmcourt <- behavior_sheet_cln |>
  ggplot(aes(nm_courtship, total_female_visits, color = average_snk)) + #gives plot dimensio
  geom_point() + #scatterplot
  geom_smooth(method ='glm.nb', se = F, color = 'darkgreen') + #Plots a Negative Binomial mo
  theme_classic()+
  labs(x = 'How many times nesting male displayed courtship behavior (count)', y = 'Females 

plot(fem.v.nmcourt)
```

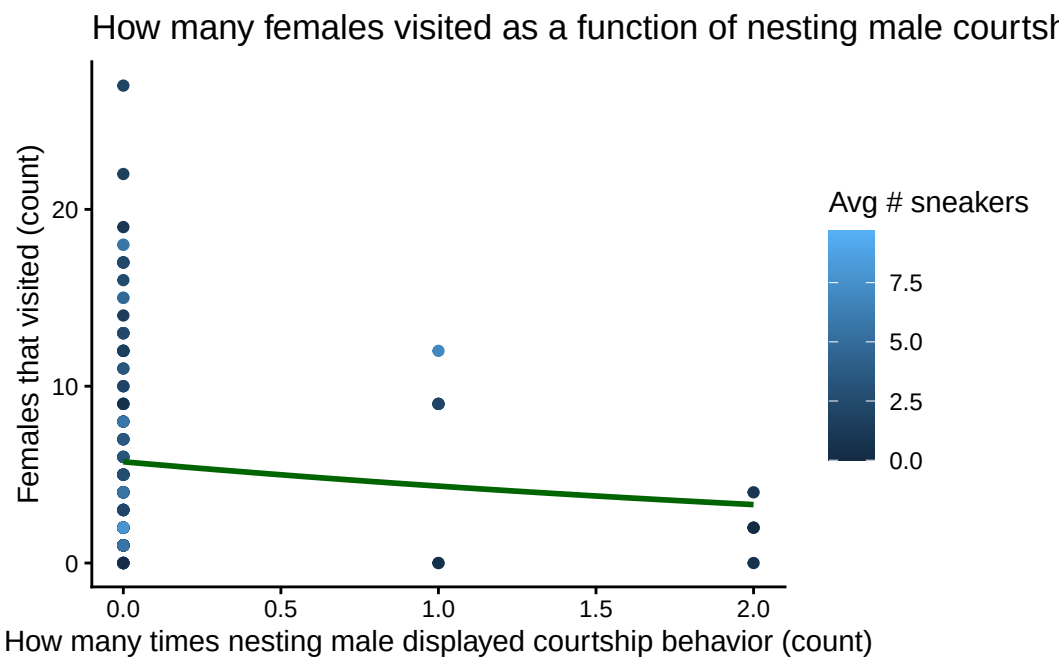
```
`geom_smooth()` using formula = 'y ~ x'
```



Total Female Spawns as a function of Nesting Male Courtship

```
fem.s.nmcourt <- behavior_sheet_cln |>
  ggplot(aes(nm_courtship, total_female_spawning, color = average_snk)) + #gives plot dimens
  geom_point() + #scatterplot
  geom_smooth(method ='glm.nb', se = F, color = 'darkgreen') + #Plots a Negative Binomial mo
  theme_classic()+
  labs(x = 'How many times nesting male displayed courtship behavior (count)', y = 'Females

plot(fem.s.nmcourt)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



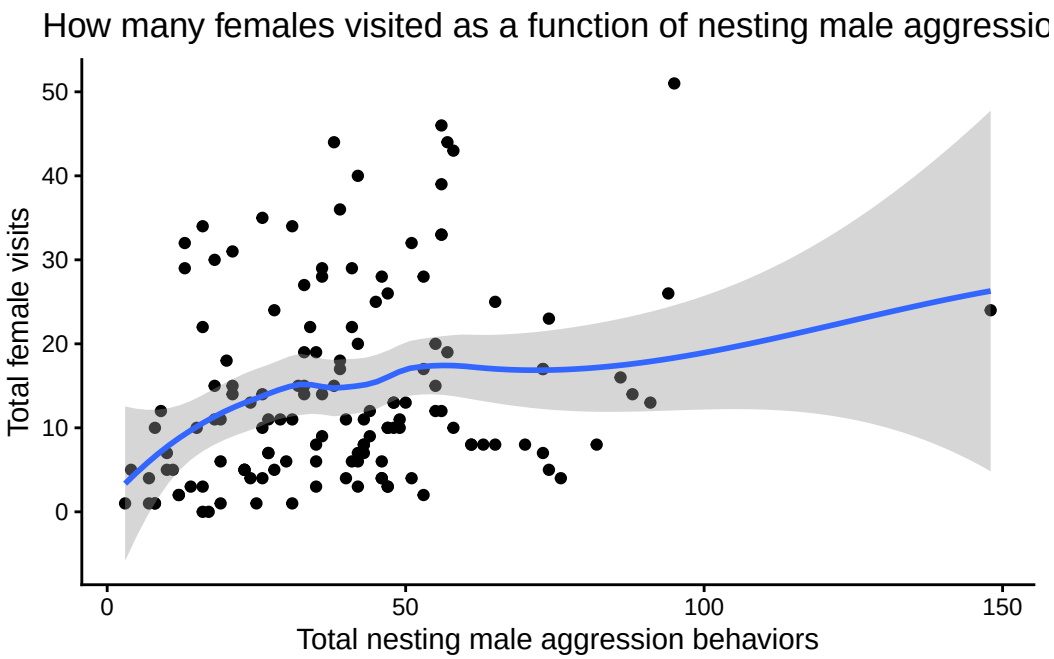
Female visits/spawning as a function of nesting male aggression

Female visits as a function of nesting male aggression

```
fvisits.nmagg <- behavior_sheet_cln |>
  #creates a variable that is a total of all NM agression
  mutate('total_nm_agg' = nm_aggression_to_sat + nm_aggression_to_snk + nm_aggression_to_othe
  #Plotting previous variable to female visits
  ggplot(aes(total_nm_agg, total_female_visits)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth() + #adds a trendline for the plot
  labs(x = 'Total nesting male aggression behaviors',
       y = 'Total female visits',
       title = 'How many females visited as a function of nesting male aggression') + #Makes
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

fvisits.nmagg
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```

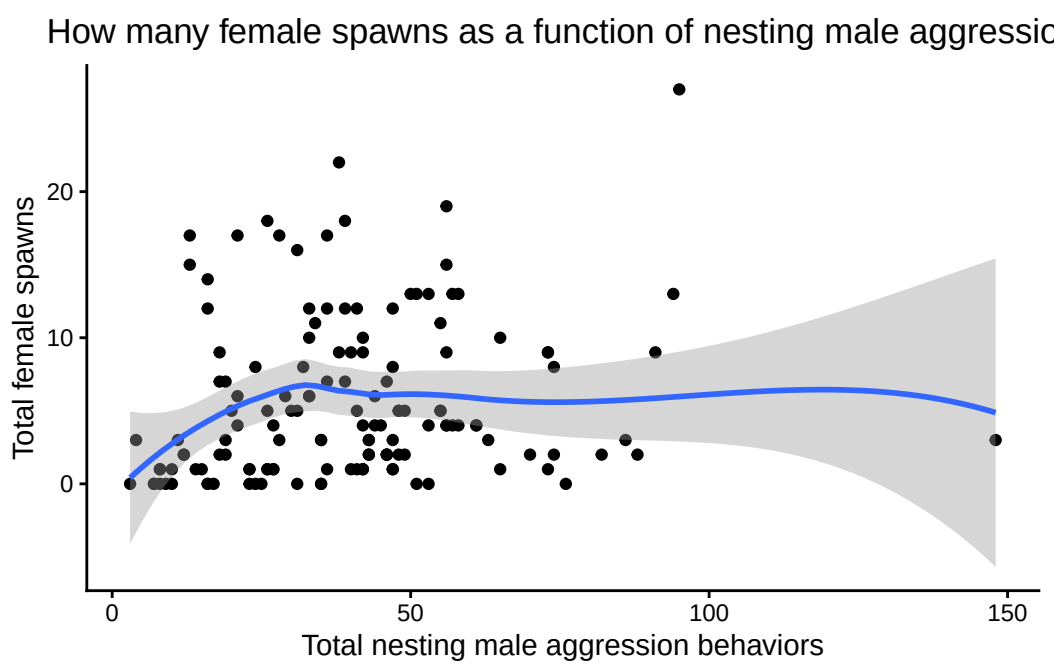


Female spawning as a function of nesting male aggression

```
f.spawns.nmagg <- behavior_sheet_cln |>
  #creates a variable that is a total of all NM agression
  mutate('total_nm_agg' = nm_aggression_to_sat + nm_aggression_to_snk + nm_aggression_to_othe
  #Plotting previous variable to female visits
  ggplot(aes(total_nm_agg, total_female_spawning)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth() + #adds a trendline for the plot
  labs(x = 'Total nesting male aggression behaviors',
       y = 'Total female spawns',
       title = 'How many female spawns as a function of nesting male aggression') + #Makes la
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle
```

```r
f.spawns.nmagg
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Models

Generalized Linear Models

Female Visits/Spawns as a function of Nesting Male Courtship

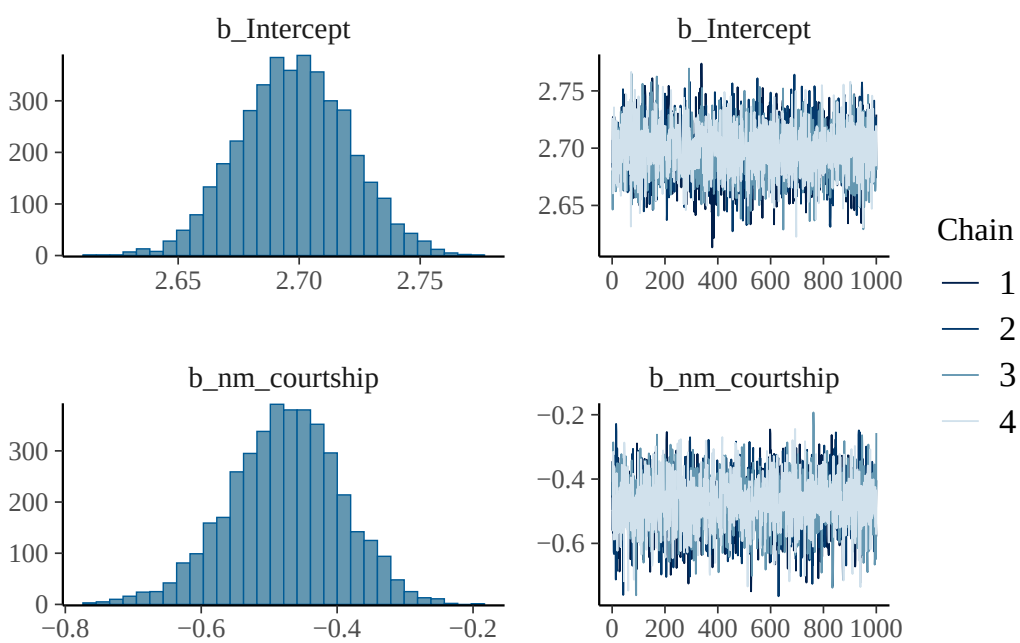
Female visits as a function of nesting male courtship

```r
fv.nm.court <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = poisson(link = 'log'), #Tell model what type of distribution
      total_female_visits ~ nm_courtship, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fv.nm.court') #saves it as an output
```

```
Compiling Stan program...
```

```
Start sampling
```

```r
print(fv.nm.court, digits=4)
plot(fv.nm.court)
```



Female spawning as a function of nesting male courtship

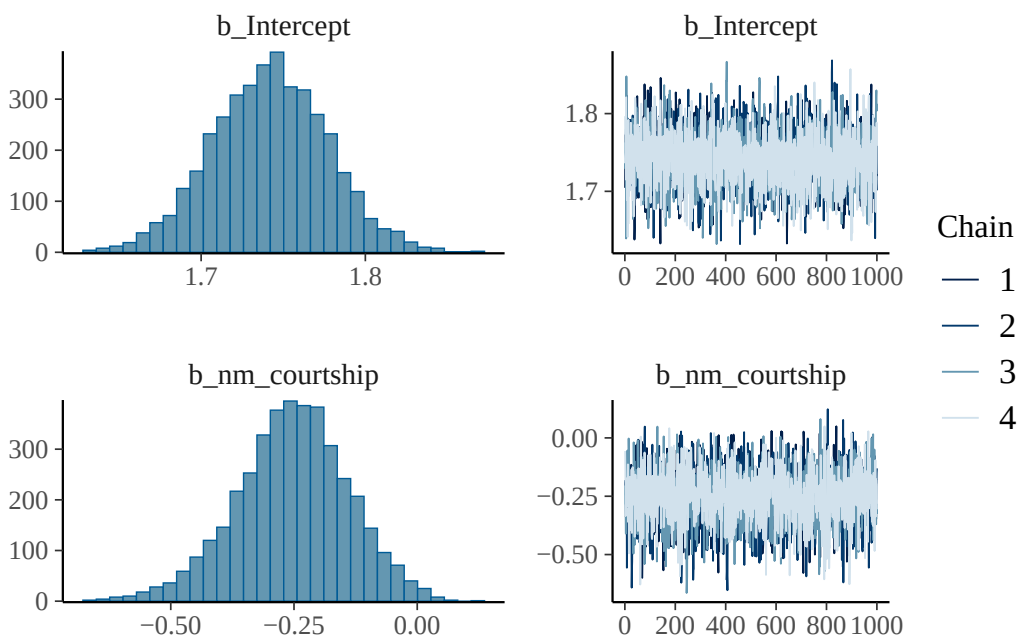
```r
fs.nm.court <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = poisson(link = 'log'), #Tell model what type of distribution
```

```
      total_female_spawning ~ nm_courtship, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fs.nm.court') #saves it as an output
```

```
Compiling Stan program...
```

```
Start sampling
```

```
print(fs.nm.court, digits=4)
plot(fs.nm.court)
```



Female visits/spawning as a function of nesting male aggression

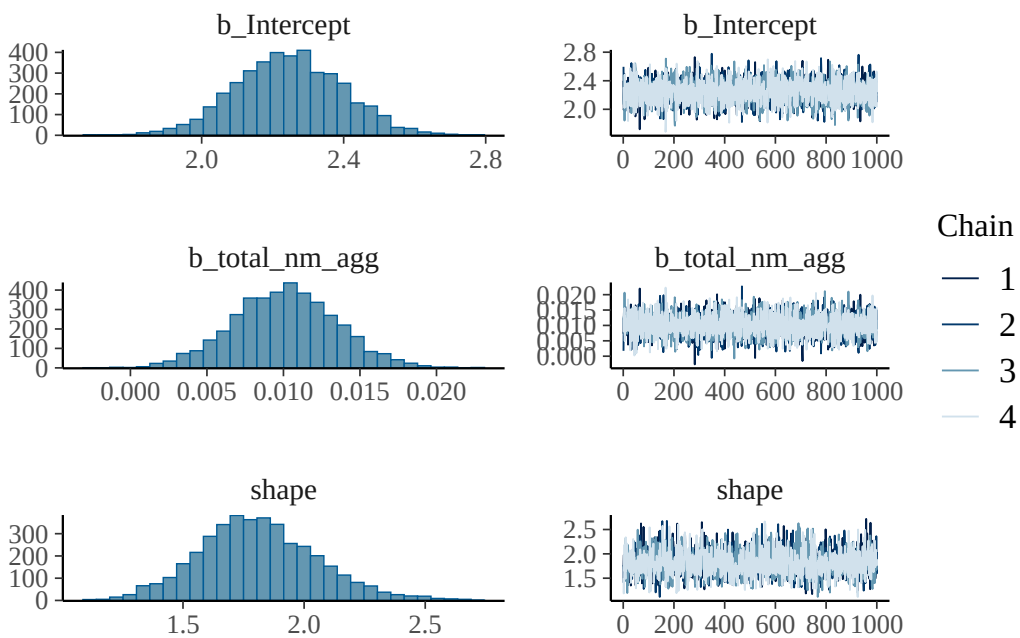
Female visits as a function of nesting male aggression

```
fvisits.nmagg <- behavior_sheet_cln |>
  #creates a variable that is a total of all NM agression
  mutate('total_nm_agg' = nm_aggression_to_sat + nm_aggression_to_snk + nm_aggression_to_othe
```

```r
fvisits.nmagg.mdl1 <-
  brm(data = fvisits.nmagg, #Give the model the tibble
      family = negbinomial(link = 'log'), #Tell model what type of distribution
      total_female_visits ~ total_nm_agg, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fvisits.nmagg.mdl1') #saves it as an output
```

```
Warning: Rows containing NAs were excluded from the model.

Compiling Stan program...

Start sampling
```

```r
print(fvisits.nmagg.mdl1, digits = 4)
plot(fvisits.nmagg.mdl1)
```

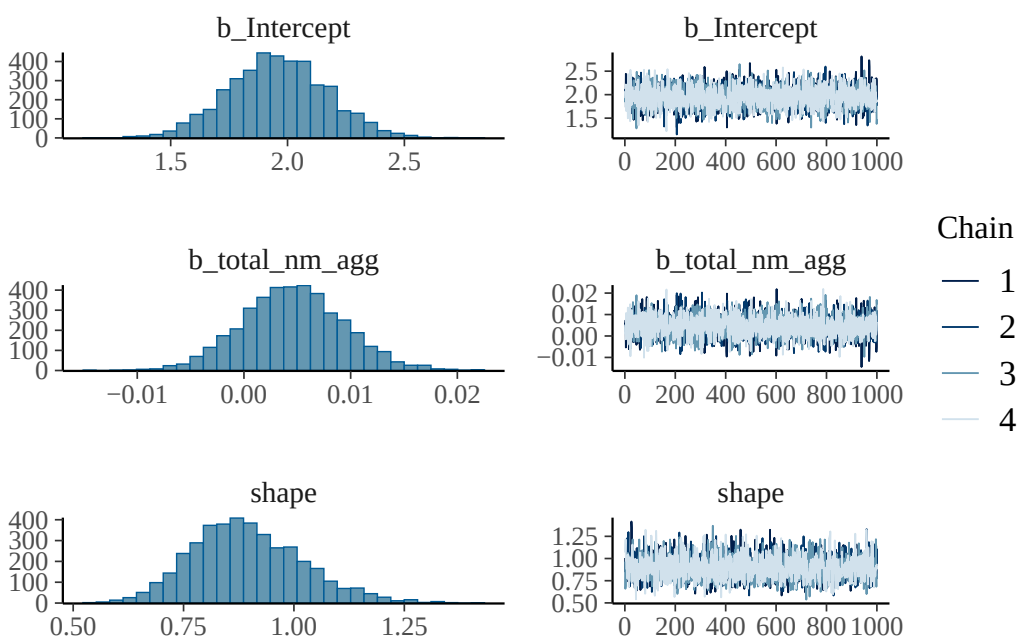


Female spawning as a function of nesting male aggression

```r
f.spawns.nmagg <- behavior_sheet_cln |>
  #creates a variable that is a total of all NM agression
  mutate('total_nm_agg' = nm_aggression_to_sat + nm_aggression_to_snk + nm_aggression_to_othe

f.spawns.nmagg.mdl1 <-
  brm(data = f.spawns.nmagg, #Give the model the tibble
      family = negbinomial(link='log'), #Tell model what type of distribution
      total_spawns ~ total_nm_agg, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/f.visits.nmagg.mdl1') #saves it as an output
```

```
Warning: Rows containing NAs were excluded from the model.

Compiling Stan program...

Start sampling
```

```r
print(f.spawns.nmagg.mdl1, digits = 4) #Gives a summary of the model
plot(f.spawns.nmagg.mdl1) #Plots the chains
```



Female spawns as a function of Her Interaction Pathway to the nest

Note:

While our data is based on counts, usually making model distributions poisson or negative binomial, we will be analyzing our data counts as binomial. This is because we are trying to see if the "interaction path", what social interaction the female had as she visited the nest, changes the probability of whether or not the female will spawn at that nest. Thus we can consider the number of female visits as the number of trials, with the probability of spawning for each interaction path being the probability.

Normalizing the pathways

By normalizing our pathways to the total number of visits, we are making each variable as a fraction of visits for each pathways, i.e. fraction of females that spawned which were brought by the SAT, fraction of females that spawned which were brought by the SAT and passed to the NM. This allows us to understand the probability of a certain pathway leading to spawning success.

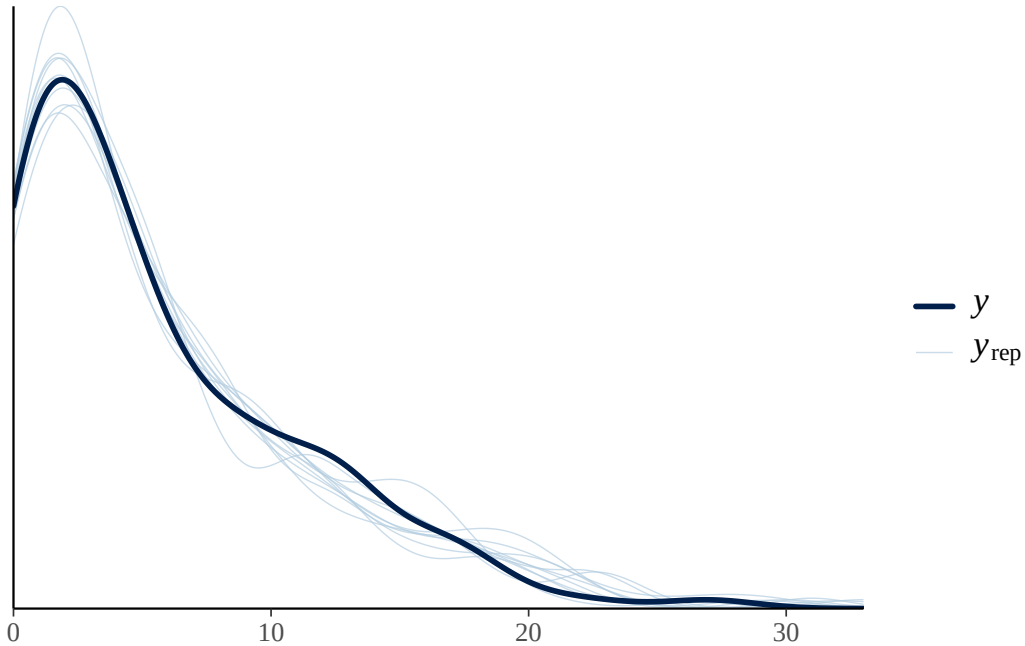
```
sheet.normalized <- behavior_sheet_cln |>
  #Normalize the different pathways by the total number of females that spawned
  mutate(
    #Nesting male brings pathway
    'fspawn.nm.b' = ((female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sat_b
    #Satellite male brings pathway
    'fspawn.sat.b' = ((female_spawning_sat_brought + females_spawing_sat_brought_sat_only_spa
    #Satellite male passes to Nesting Male pathway
    'fspawn.sat.b.p.nm' = ((female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to_
  )
```

Model 1 Just Nesting Male

```
sheet.normalized.rmv <- sheet.normalized |>
  #Removing rows where there were data mistakes
  filter(!(total_female_spawning > total_female_visits))


fem.spawns.path.mdl.1 <- sheet.normalized.rmv |>
  brm(data = sheet.normalized.rmv,
    family = binomial(link = 'logit'), #Tell model what type of distribution
    #Specifying the model
    formula = bf(total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.nm.b),
```

```r
    prior = c(prior(normal(0, 2), class = Intercept)),
    iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fem.spawns.path.mdl.1') #saves it as an output

#Gives a plot of the model's predictions
pp_check(fem.spawns.path.mdl.1)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

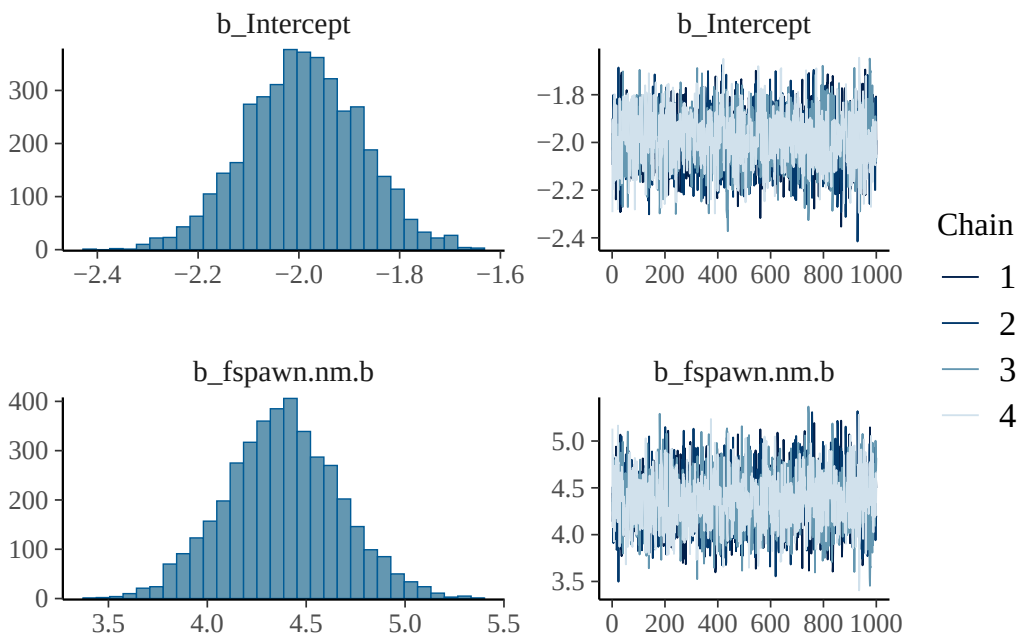



```r
#Gives us the summary of the model
print(fem.spawns.path.mdl.1, digits = 4)
```

```
 Family: binomial
  Links: mu = logit
Formula: total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.nm.b
         autocor ~ tructure(list(), class = "formula", .Environment = <environment>)
   Data: sheet.normalized.rmv (Number of observations: 137)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000
```

```
Regression Coefficients:
            Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
Intercept    -1.9908    0.1143  -2.2175  -1.7705 1.0014     2047     2454
fspawn.nm.b   4.3802    0.2892   3.8231   4.9619 1.0004     2248     2442

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#Plots the Markov chains
plot(fem.spawns.path.mdl.1)
```

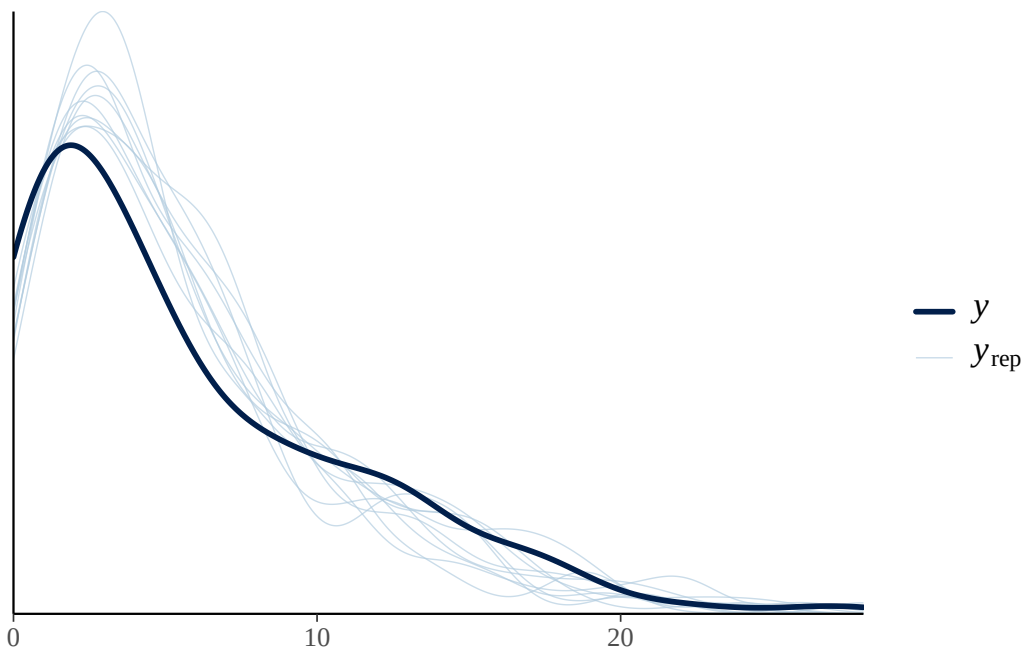


Model 2 Just Satellite

```r
sheet.normalized.rmv <- sheet.normalized |>
  #Removing rows where there were data mistakes
  filter(!(total_female_spawning > total_female_visits))


fem.spawns.path.mdl.2 <- sheet.normalized.rmv |>
  brm(data = sheet.normalized.rmv,
    family = binomial(link = 'logit'), #Tell model what type of distribution
```

```r
    #Specifying the model
    formula = bf(total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.sat.b),
    prior = c(prior(normal(0, 2), class = Intercept)),
    iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fem.spawns.path.mdl.2') #saves it as an output

#Gives a plot of the model's predictions
pp_check(fem.spawns.path.mdl.2)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

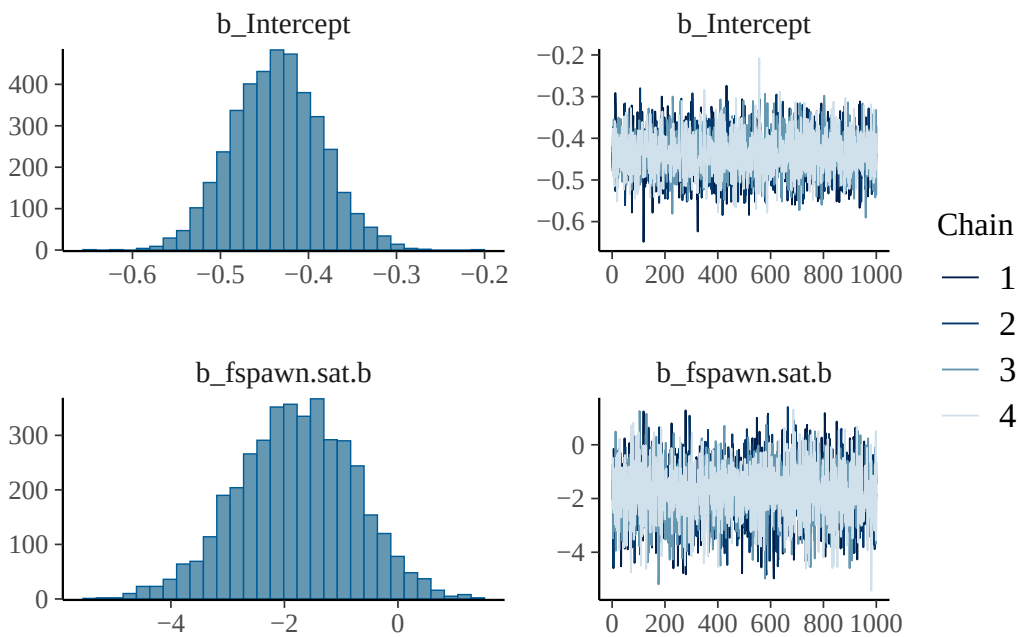



```r
#Gives us the summary of the model
print(fem.spawns.path.mdl.2, digits = 4)
```

```
 Family: binomial
  Links: mu = logit
Formula: total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.sat.b
         autocor ~ tructure(list(), class = "formula", .Environment = <environment>)
   Data: sheet.normalized.rmv (Number of observations: 137)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
```

```
         total post-warmup draws = 4000

Regression Coefficients:
             Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
Intercept     -0.4362    0.0506  -0.5335  -0.3354 1.0006     3218     2376
fspawn.sat.b  -1.7931    1.0320  -3.8757   0.1840 1.0010     3352     2564

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#Plots the Markov chains
plot(fem.spawns.path.mdl.2)
```

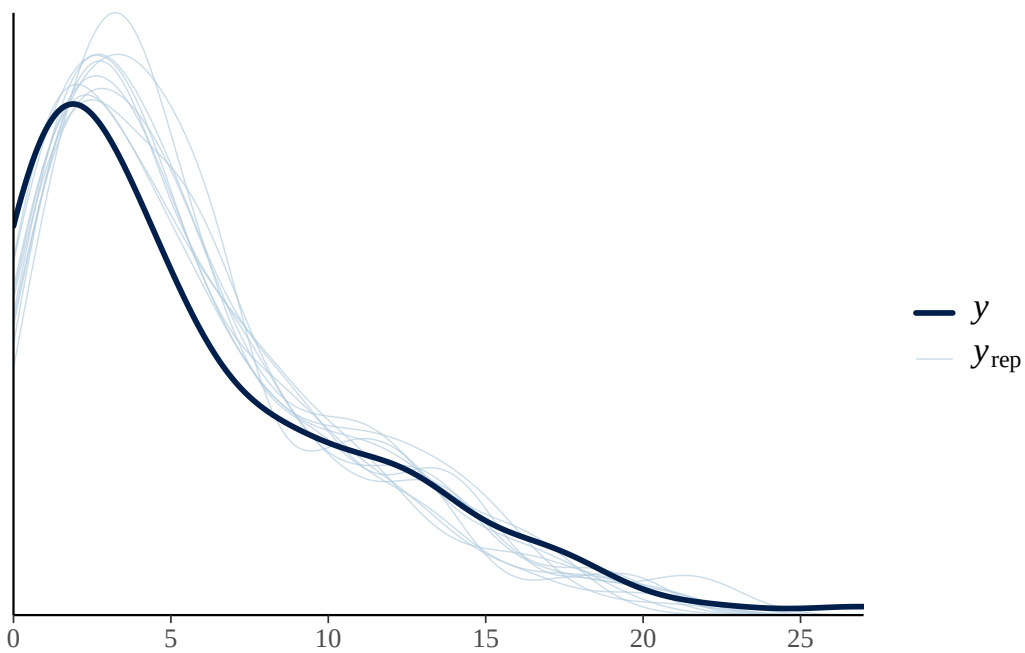


Model 3 Just Satellite pass to Nesting Male

```r
sheet.normalized.rmv <- sheet.normalized |>
  #Removing rows where there were data mistakes
  filter(!(total_female_spawning > total_female_visits))


fem.spawns.path.mdl.3 <- sheet.normalized.rmv |>
```

```r
  brm(data = sheet.normalized.rmv,
    family = binomial(link = 'logit'), #Tell model what type of distribution
    #Specifying the model
    formula = bf(total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.sat.b.p.nm
    prior = c(prior(normal(0, 2), class = Intercept)),
    iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fem.spawns.path.mdl.3') #saves it as an output

#Gives a plot of the model's predictions
pp_check(fem.spawns.path.mdl.3)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

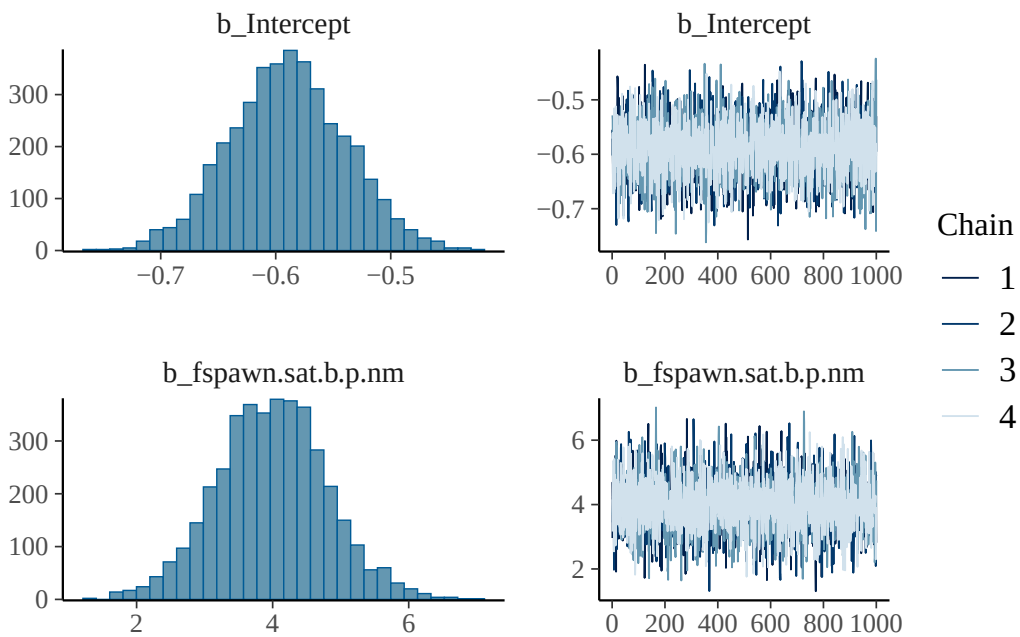



```r
#Gives us the summary of the model
print(fem.spawns.path.mdl.3, digits = 4)
```

```
 Family: binomial
  Links: mu = logit
Formula: total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.sat.b.p.nm
         autocor ~ tructure(list(), class = "formula", .Environment = <environment>)
```

```
   Data: sheet.normalized.rmv (Number of observations: 137)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                  Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
Intercept          -0.5894    0.0506  -0.6885  -0.4897 0.9997     3645     2734
fspawn.sat.b.p.nm   3.9888    0.8089   2.3920   5.6370 1.0002     3659     2585

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
#Plots the Markov chains
plot(fem.spawns.path.mdl.3)
```

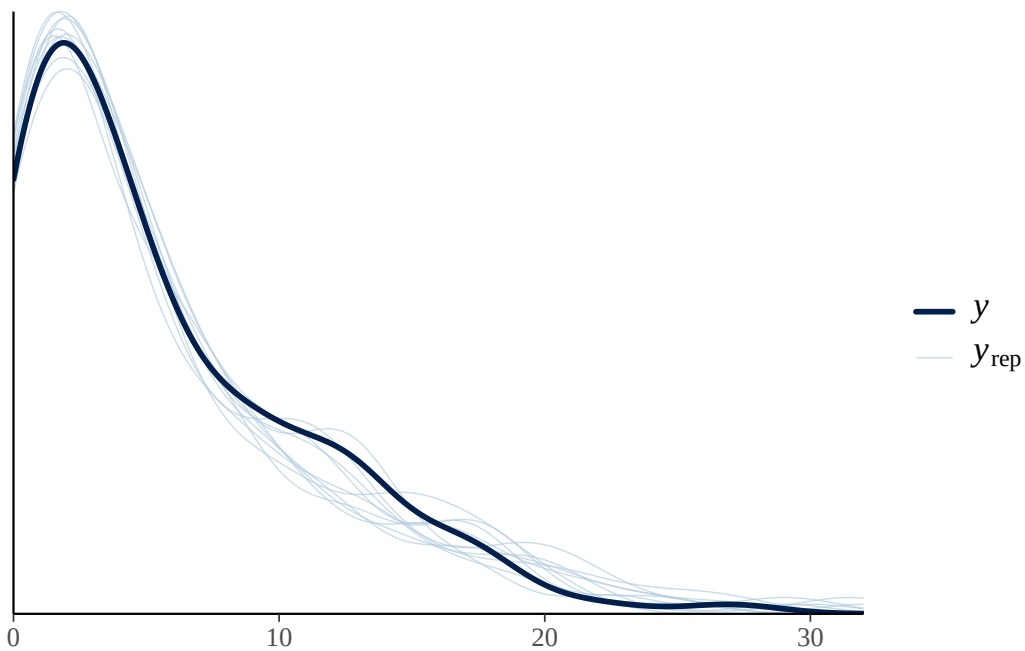


Model 4, Just Nesting Male and Just Satellite

```
sheet.normalized.rmv <- sheet.normalized |>
  #Removing rows where there were data mistakes
  filter(!(total_female_spawning > total_female_visits))
```

```r
fem.spawns.path.mdl.4 <- sheet.normalized.rmv |>
  brm(data = sheet.normalized.rmv,
    family = binomial(link = 'logit'), #Tell model what type of distribution
    #Specifying the model
    formula = bf(total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.nm.b + fspa
    prior = c(prior(normal(0, 2), class = Intercept)),
    iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fem.spawns.path.mdl.4') #saves it as an output

#Gives a plot of the model's predictions
pp_check(fem.spawns.path.mdl.4)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

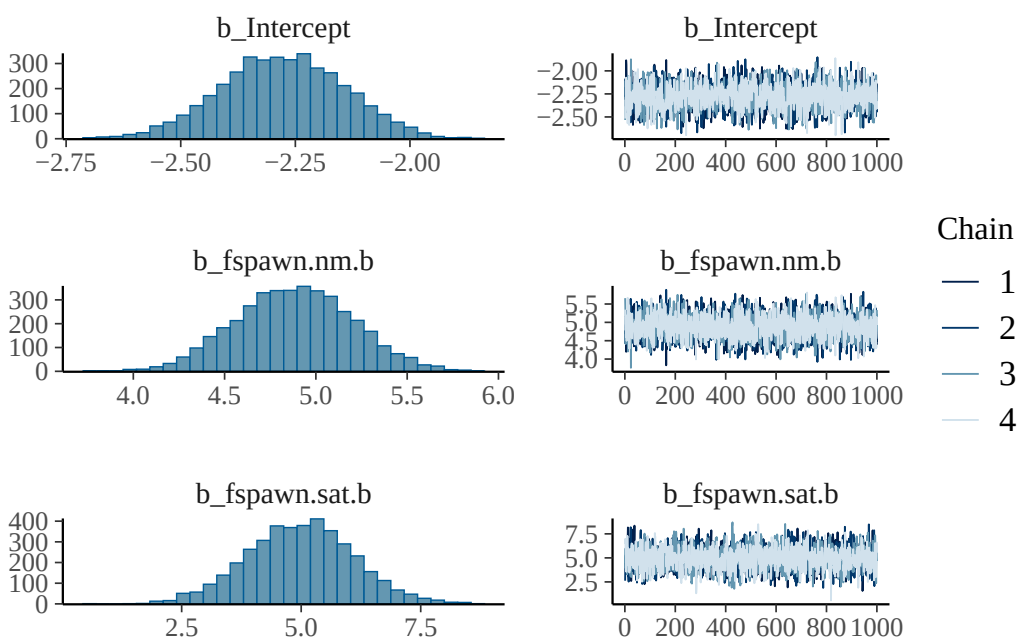



```r
#Gives us the summary of the model
print(fem.spawns.path.mdl.4, digits = 4)
```

```
 Family: binomial
  Links: mu = logit
```

```
Formula: total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.nm.b + fspawn.sat.b
         autocor ~ tructure(list(), class = "formula", .Environment = <environment>)
   Data: sheet.normalized.rmv (Number of observations: 137)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
             Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
Intercept     -2.2776    0.1348  -2.5449  -2.0185 1.0011     2184     2387
fspawn.nm.b    4.8829    0.3167   4.2816   5.5064 1.0006     2585     2685
fspawn.sat.b   4.9825    1.1106   2.7546   7.1820 1.0006     2831     2569

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

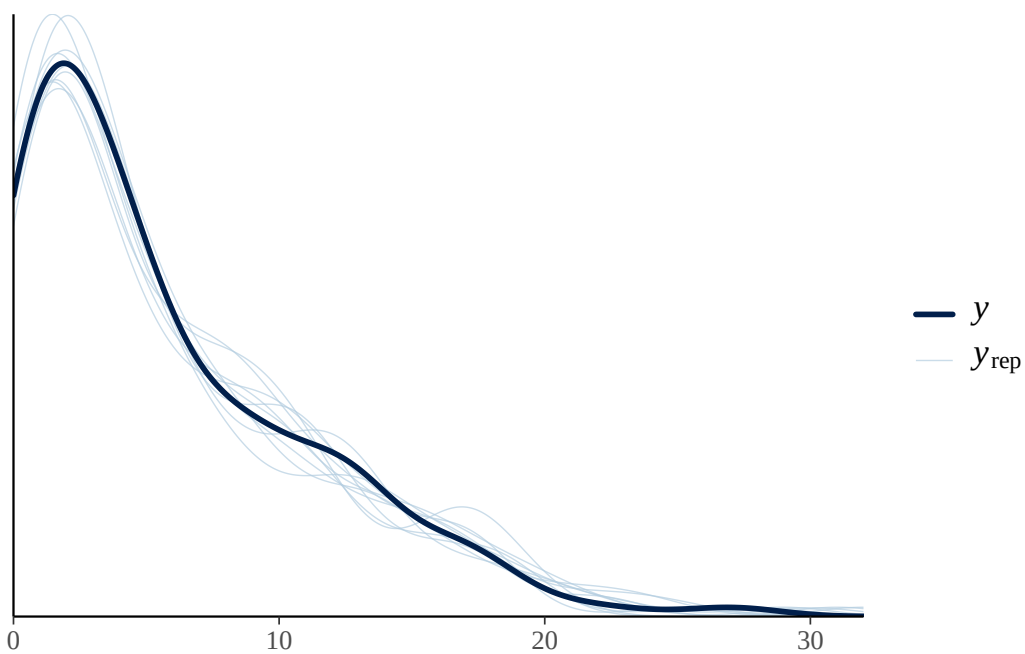
```r
#Plots the Markov chains
plot(fem.spawns.path.mdl.4)
```



Model 5, Just Nesting Male and Just Satellite pass to Nesting Male

```r
sheet.normalized.rmv <- sheet.normalized |>
  #Removing rows where there were data mistakes
  filter(!(total_female_spawning > total_female_visits))


fem.spawns.path.mdl.5 <- sheet.normalized.rmv |>
  brm(data = sheet.normalized.rmv,
    family = binomial(link = 'logit'), #Tell model what type of distribution
    #Specifying the model
    formula = bf(total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.nm.b + fspa
    prior = c(prior(normal(0, 2), class = Intercept)),
    iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fem.spawns.path.mdl.5') #saves it as an output

#Gives a plot of the model's predictions
pp_check(fem.spawns.path.mdl.5)
```

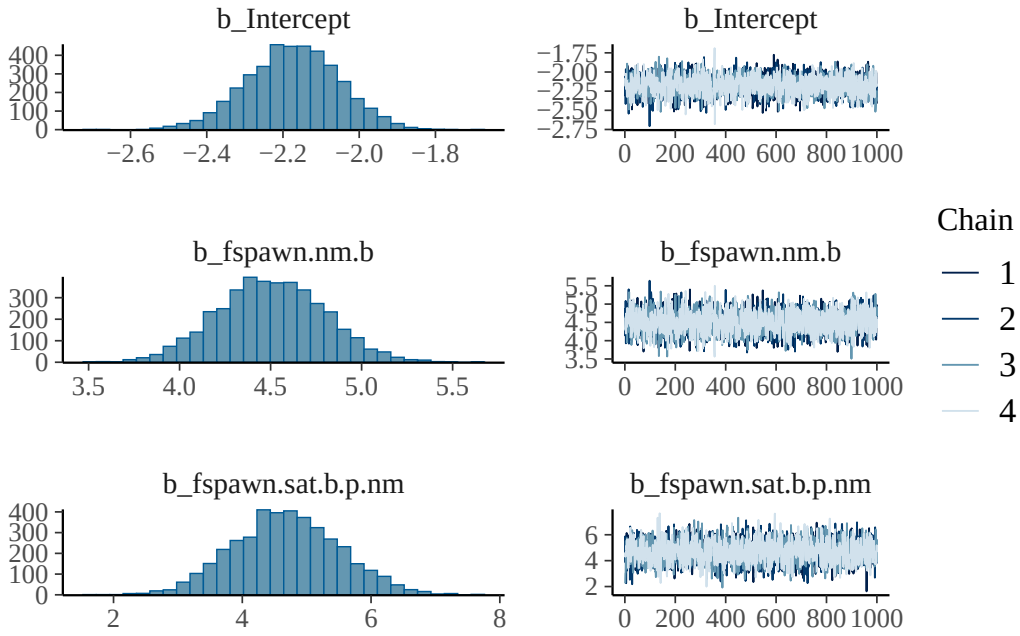
```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```



```r
#Gives us the summary of the model
print(fem.spawns.path.mdl.5, digits = 4)
```

```
 Family: binomial
  Links: mu = logit
Formula: total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.nm.b + fspawn.sat.b
         autocor ~ tructure(list(), class = "formula", .Environment = <environment>)
   Data: sheet.normalized.rmv (Number of observations: 137)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                  Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
Intercept          -2.1742    0.1206  -2.4138  -1.9456 1.0004     2959     2852
fspawn.nm.b         4.5036    0.2927   3.9434   5.0828 1.0007     3777     3336
fspawn.sat.b.p.nm   4.6878    0.8239   3.1239   6.3123 1.0008     3462     2876

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

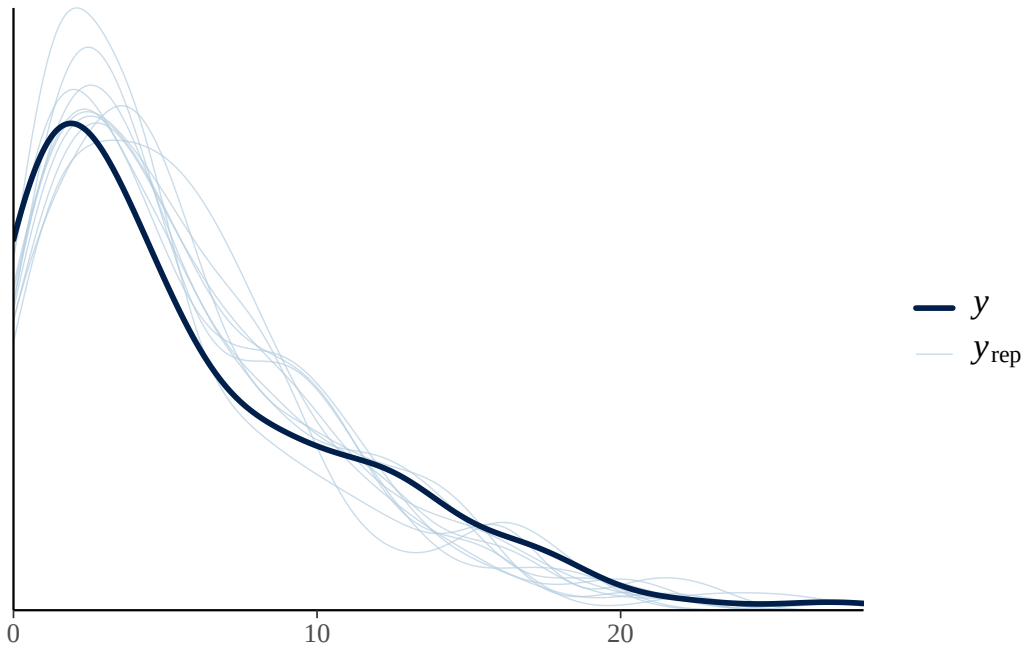
```r
#Plots the Markov chains
plot(fem.spawns.path.mdl.5)
```



Model 6, Just Satellite and Just Satellite pass to Nesting Male

```
sheet.normalized.rmv <- sheet.normalized |>
  #Removing rows where there were data mistakes
  filter(!(total_female_spawning > total_female_visits))


fem.spawns.path.mdl.6 <- sheet.normalized.rmv |>
  brm(data = sheet.normalized.rmv,
    family = binomial(link = 'logit'), #Tell model what type of distribution
    #Specifying the model
    formula = bf(total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.sat.b + fsp
    prior = c(prior(normal(0, 2), class = Intercept)),
    iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/fem.spawns.path.mdl.6') #saves it as an output


#Gives a plot of the model's predictions
pp_check(fem.spawns.path.mdl.6)
```

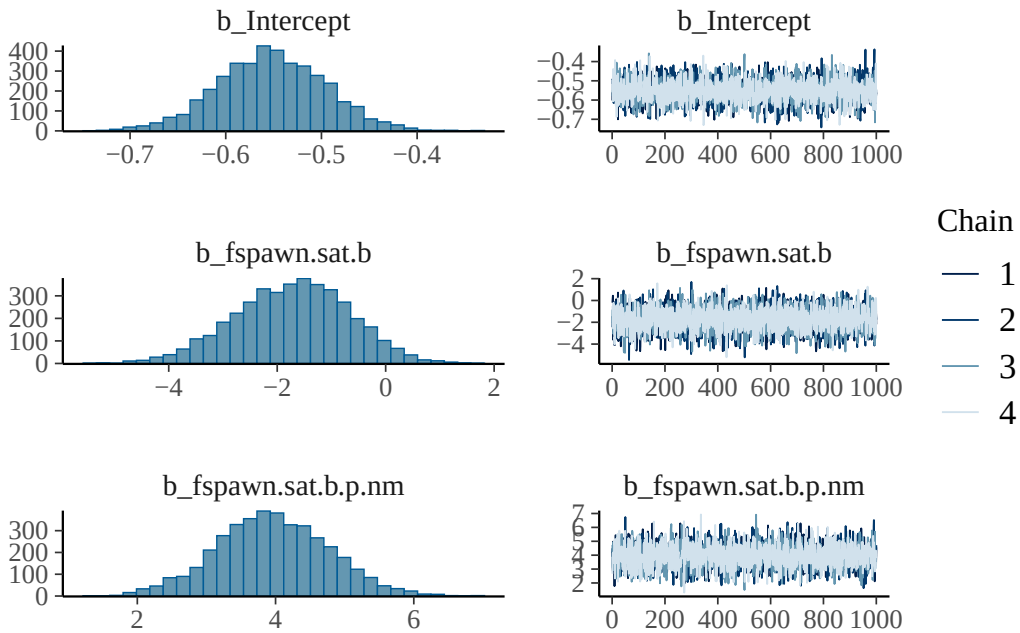
```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```




```r
#Gives us the summary of the model
print(fem.spawns.path.mdl.6, digits = 4)
```

```
 Family: binomial
  Links: mu = logit
Formula: total_female_spawning | trials(total_female_visits) ~ 1 + fspawn.sat.b + fspawn.sat
         autocor ~ tructure(list(), class = "formula", .Environment = <environment>)
   Data: sheet.normalized.rmv (Number of observations: 137)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                  Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
Intercept          -0.5522    0.0564  -0.6631  -0.4413 1.0005     4844     3035
fspawn.sat.b       -1.7256    1.0513  -3.8287   0.2389 1.0020     4518     2990
fspawn.sat.b.p.nm   3.9597    0.8091   2.3769   5.5587 1.0006     4213     2935

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#Plots the Markov chains
plot(fem.spawns.path.mdl.6)
```



Compare the models

```r
waic(fem.spawns.path.mdl.1)
```

```
Computed from 4000 by 137 log-likelihood matrix.

          Estimate   SE
elpd_waic   -202.6  9.3
p_waic         1.0  0.2
waic         405.3 18.5
```

```r
waic(fem.spawns.path.mdl.2)
```

```
Warning:
1 (0.7%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```
Computed from 4000 by 137 log-likelihood matrix.

          Estimate   SE
elpd_waic   -346.1 19.2
p_waic         5.3  1.6
waic         692.3 38.5

1 (0.7%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```r
waic(fem.spawns.path.mdl.3)
```

```
Warning:
1 (0.7%) p_waic estimates greater than 0.4. We recommend trying loo instead.


Computed from 4000 by 137 log-likelihood matrix.

          Estimate   SE
elpd_waic   -334.6 18.7
p_waic         4.4  0.9
waic         669.1 37.4

1 (0.7%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```r
waic(fem.spawns.path.mdl.4)
```

```
Computed from 4000 by 137 log-likelihood matrix.

          Estimate   SE
elpd_waic   -193.2  9.2
p_waic         1.1  0.4
waic         386.3 18.5
```

```r
waic(fem.spawns.path.mdl.5)
```

```
Computed from 4000 by 137 log-likelihood matrix.
```

```
          Estimate   SE
elpd_waic   -187.3  7.5
p_waic         0.7  0.2
waic         374.5 15.1
```

```r
waic(fem.spawns.path.mdl.6)
```

```
Warning:
2 (1.5%) p_waic estimates greater than 0.4. We recommend trying loo instead.


Computed from 4000 by 137 log-likelihood matrix.

          Estimate   SE
elpd_waic   -335.4 18.4
p_waic         7.3  1.9
waic         670.9 36.9

2 (1.5%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

Based on the waic values, model 5 which is our model with the Nesting male bringing females and Satellite passing to Nesting male pathways is the best model given our data. This makes biological sense, the better a Nesting male and Satellite pair are the more likely a nest is going to be efficient, chasing sneakers and bringing females!

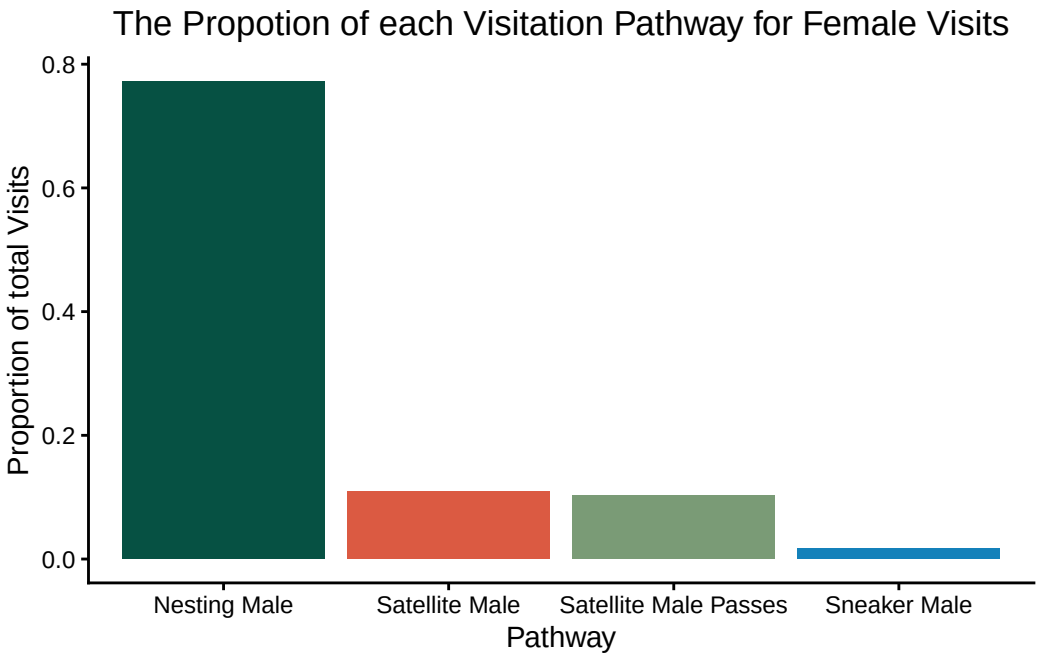
More Aligned Models

While the previous model is interesting, it isn't quite telling us what we want to know. Our previous model was inferring that the total number of females spawning were best predicted by the females that spawned when the nesting male brought a female or when a Satellite passed the female to the nesting male. What we really want to know is if there is a difference in female spawning decision based on which 'pathway' she was introduced to the nest. Thus we need to make the model a little more complex.

Goal: Given how a female visits, does it change the likelihood of spawning (in each pathway). Does that then correlate to overall nest success? Are there other influencing factors (number of sneakers, +/- nesting male and satellite male interactions?)

Let's visualize the pathways.

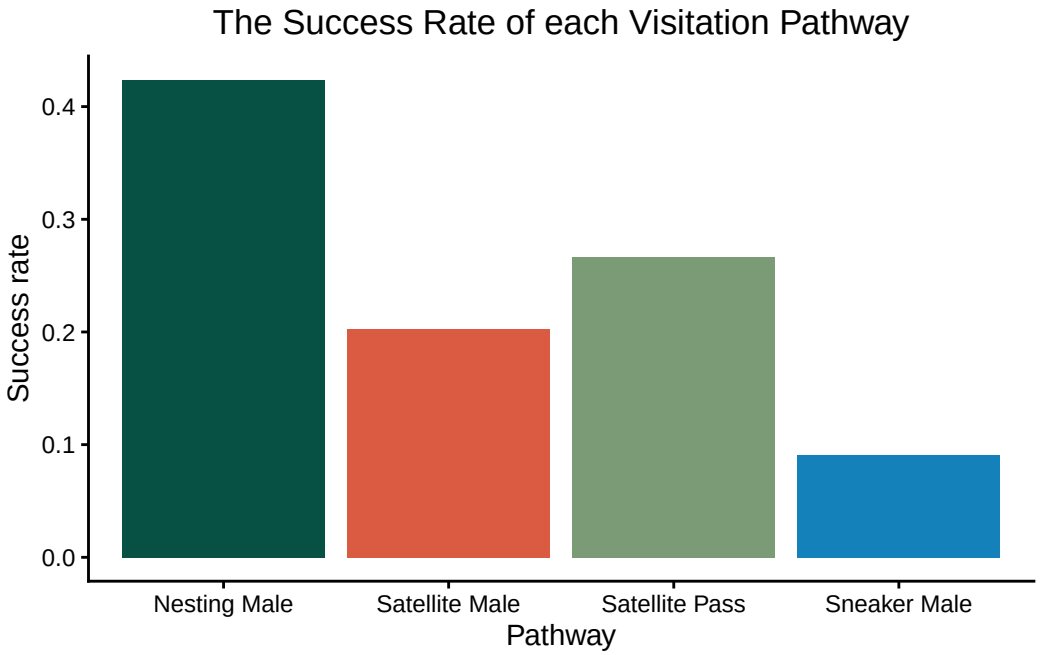
```r
pathways.visits <- behavior_sheet_cln |>
    #Remove nests where there are data inconsistencies
  filter_out(female_visits_by_nm > total_female_visits) |>
  filter_out(female_visit_by_sat > total_female_visits) |>
  filter_out(female_visits_by_sat_passed_to_nm > total_female_visits)|>
  filter_out(female_visit_by_snk > total_female_visits)|>
  #Create new variables that account for the pathway's proportion
  mutate(
    'Nesting Male' = sum(female_visits_by_nm)/sum(total_female_visits),
    'Satellite Male' = sum(female_visit_by_sat)/sum(total_female_visits),
    'Satellite Male Passes' = sum(female_visits_by_sat_passed_to_nm)/sum(total_female_visits
    'Sneaker Male' = sum(female_visit_by_snk)/sum(total_female_visits)
  ) |>
  #Make the tibble only include these variables
  dplyr::select(`Nesting Male`:`Sneaker Male`) |>
  #Remove all the duplicates
  distinct() |>
  #Adjust the table to make the variables go into one coloumn
  pivot_longer(`Nesting Male`:`Sneaker Male`,
               names_to = 'Pathway',
               values_to = 'Proportion') |>
  #Graph it
  ggplot(aes(Pathway, Proportion, fill = Pathway)) +
  #Makes the graph a bar graph
  geom_col() +
  labs(x = 'Pathway',
       y = 'Proportion of total Visits',
       title = "The Propotion of each Visitation Pathway for Female Visits",) + #edits axis a
  theme_classic() + #Sets the plot visuals
  #Adds a color scheme to the plot
  scale_fill_manual(values = c('Nesting Male' = '#065143', 'Satellite Male' = '#db5a42', 'Sat
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  theme(legend.position = "none") # Removes all legends

pathways.visits
```



Now let's plot the success rate of each pathway:

```
pathways.success.rate <- behavior_sheet_cln |>
  #Remove nests where there are data inconsistencies
  filter_out((female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sat_brought_b
  filter_out((female_spawning_sat_brought + females_spawing_sat_brought_sat_only_spawn)/(fema
  filter_out((female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to_nm_but_only_sa
  filter_out((female_spawning_snk_brought + female_spawning_snk_brought_skn_only_spawn)/(fema
  #Create new variables that account for the pathway's proportion
  mutate(
    'Nesting Male' = sum(female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sa
    'Satellite Male' = sum(female_spawning_sat_brought + females_spawing_sat_brought_sat_only
    'Satellite Pass' = sum(female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to_n
    'Sneaker Male' = sum(female_spawning_snk_brought + female_spawning_snk_brought_skn_only_s
  ) |>
  #Make the tibble only include these variables
  dplyr::select(`Nesting Male`:`Sneaker Male`) |>
  #Remove all the duplicates
  distinct() |>
  #Adjust the table to make the variables go into one coloumn
  pivot_longer(`Nesting Male`:`Sneaker Male`,
               names_to = 'Pathway',
               values_to = 'Success Rate') |>
```

```
  #Graph it
  ggplot(aes(Pathway, `Success Rate`, fill = Pathway)) +
  #Makes the graph a bar graph
  geom_col() +
  labs(x = 'Pathway',
       y = 'Success rate',
       title = "The Success Rate of each Visitation Pathway",) + #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  #Adds a color scheme to the plot
  scale_fill_manual(values = c('Nesting Male' = '#065143', 'Satellite Male' = '#db5a42', 'Sat
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  theme(legend.position = "none") # Removes all legends

pathways.success.rate
```

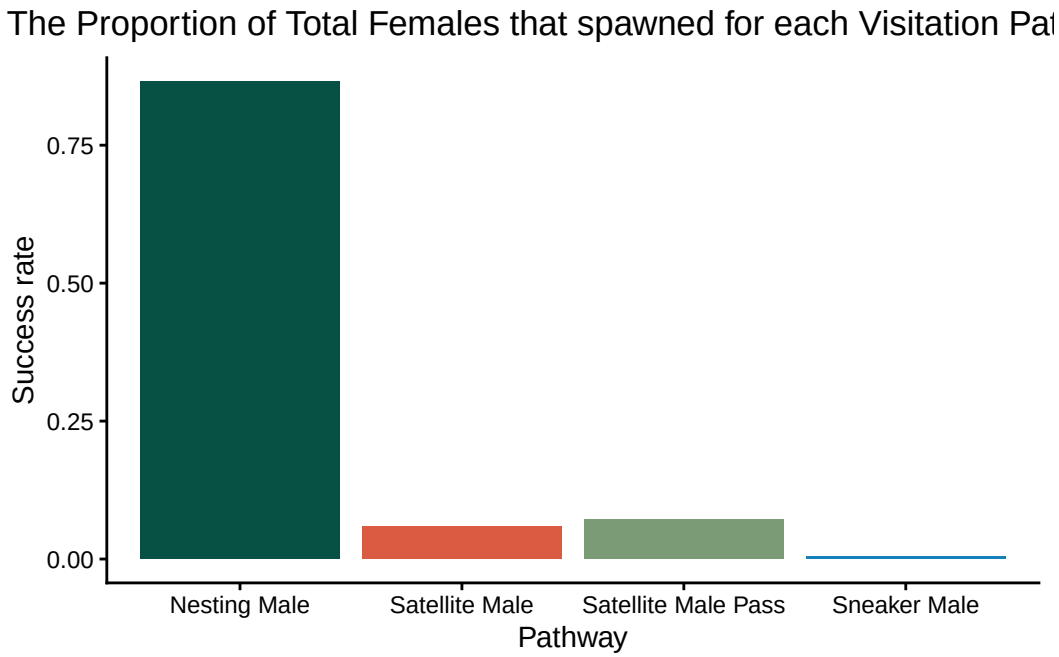


The presence of the sneaker male rate could be because of secondary satellites being counted as sneakers

Now let's visualize how often these pathway successes occur in total at a nest.

```
pathways.success.rate.total <- behavior_sheet_cln |>
  #Remove nests where there are data inconsistencies
  filter_out((female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sat_brought_
```

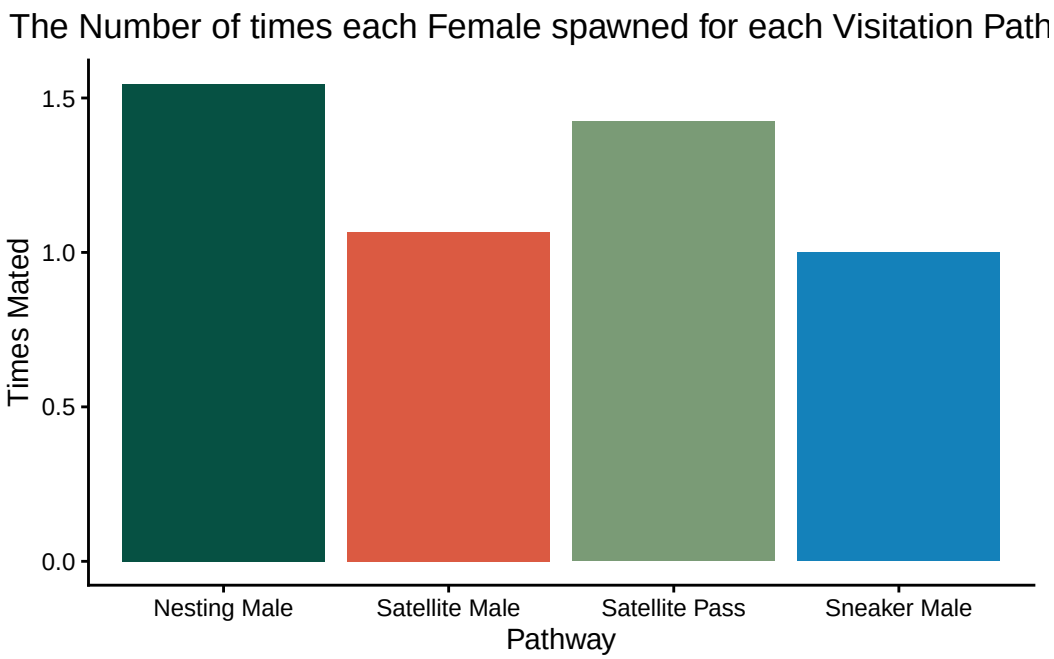
```
  filter_out((female_spawning_sat_brought + females_spawing_sat_brought_sat_only_spawn)/(tota
  filter_out((female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to_nm_but_only_sa
  filter_out((female_spawning_snk_brought + female_spawning_snk_brought_skn_only_spawn)/(tota
  #Create new variables that account for the pathway's proportion
  mutate(
    'Nesting Male' = sum(female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sa
    'Satellite Male' = sum(female_spawning_sat_brought + females_spawing_sat_brought_sat_only
    'Satellite Male Pass' = sum(female_spawning_sat_passed_to_nm + female_spawning_sat_passed
    'Sneaker Male' = sum(female_spawning_snk_brought + female_spawning_snk_brought_skn_only_s
  ) |>
  #Make the tibble only include these variables
  dplyr::select(`Nesting Male`:`Sneaker Male`) |>
  #Remove all the duplicates
  distinct() |>
  #Adjust the table to make the variables go into one coloumn
  pivot_longer(`Nesting Male`:`Sneaker Male`,
               names_to = 'Pathway',
               values_to = 'Success Rate') |>
  #Graph it
  ggplot(aes(Pathway, `Success Rate`, fill = Pathway)) +
  #Makes the graph a bar graph
  geom_col() +
  labs(x = 'Pathway',
       y = 'Success rate',
       title = "The Proportion of Total Females that spawned for each Visitation Pathway",) +
  theme_classic() + #Sets the plot visuals
  #Adds a color scheme to the plot
  scale_fill_manual(values = c('Nesting Male' = '#065143', 'Satellite Male' = '#db5a42', 'Sat
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  theme(legend.position = "none") # Removes all legends

pathways.success.rate.total
```



Let's visualize if certain pathways lead to females spawning more

```
pathways.success.rate.total <- behavior_sheet_cln |>
  #Remove nests where there are data inconsistencies
  filter_out((female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sat_brought_b
  filter_out((female_spawning_sat_brought + females_spawing_sat_brought_sat_only_spawn) > (to
  filter_out((female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to_nm_but_only_sa
  filter_out((female_spawning_snk_brought + female_spawning_snk_brought_skn_only_spawn) > (to
  #Create new variables that account for the pathway's proportion
  mutate(
    'Nesting Male' = sum(total_spawns_nm_brought + total_spawns_nm_brought_sat_only_spawn)/su
    'Satellite Male' = sum(total_spawns_sat_brought + total_spawns_sat_brought_sat_only_spawn
    'Satellite Pass' = sum(total_spawns_sat_passed_to_nm + total_spawns_sat_passed_to_nm_but_
    'Sneaker Male' = sum(total_spawns_snk_brought + total_spawns_snk_brought_snk_only_spawn)/
  ) |>
  #Make the tibble only include these variables
  dplyr::select(`Nesting Male`:`Sneaker Male`) |>
  #Remove all the duplicates
  distinct() |>
  #Adjust the table to make the variables go into one coloumn
  pivot_longer(`Nesting Male`:`Sneaker Male`,
               names_to = 'Pathway',
               values_to = 'Success Rate') |>
```

```
  #Graph it
  ggplot(aes(Pathway, `Success Rate`, fill = Pathway)) +
  #Makes the graph a bar graph
  geom_col() +
  labs(x = 'Pathway',
       y = 'Times Mated',
       title = "The Number of times each Female spawned for each Visitation Pathway",) + #edi
  theme_classic() + #Sets the plot visuals
  #Adds a color scheme to the plot
  scale_fill_manual(values = c('Nesting Male' = '#065143', 'Satellite Male' = '#db5a42', 'Sat
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  theme(legend.position = "none") # Removes all legends

pathways.success.rate.total
```



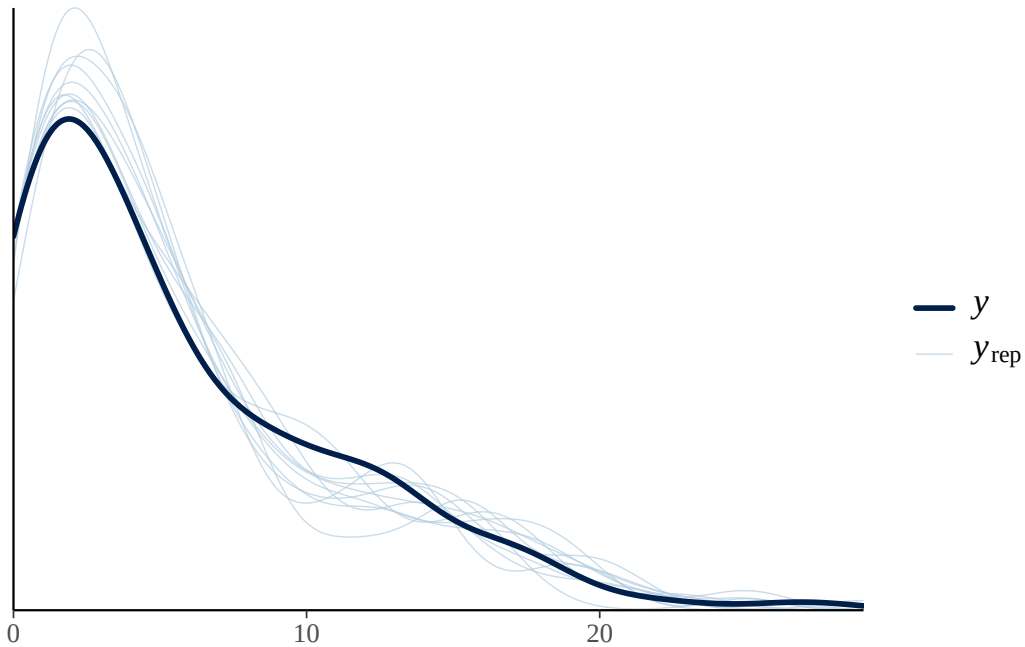
Now let's run some models

We'll run the Nesting Male and Satellite pass to Nesting Male pathways first since it seemed the best fit before.

```
path.visits.norm <- behavior_sheet_cln |>
  #Normalize the different pathways by the total number of females that visited
  mutate(
```

```
    'nm.path' = (female_visits_by_nm)/(total_female_visits),
    'sat.path' = (female_visit_by_sat)/(total_female_visits),
    'sat.pass.path' = (female_visits_by_sat_passed_to_nm)/(total_female_visits),
    'snk.path' = (female_visit_by_snk)/(total_female_visits)
  )

path.visits.norm.rmv <- path.visits.norm |>
  #Removing rows where there were data mistakes
  filter(!(total_female_spawning > total_female_visits))


path.fem.spawning <- path.visits.norm.rmv |>
  brm(data = path.visits.norm.rmv,
    family = binomial(link = 'logit'), #Tell model what type of distribution
    #Specifying the model
    formula = bf(total_female_spawning | trials(total_female_visits) ~ 1 + nm.path + sat.pass
    prior = c(prior(normal(0, 2), class = Intercept)),
    iter = 4000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/path.fem.spawning') #saves it as an output

#Gives a plot of the model's predictions
pp_check(path.fem.spawning)
```

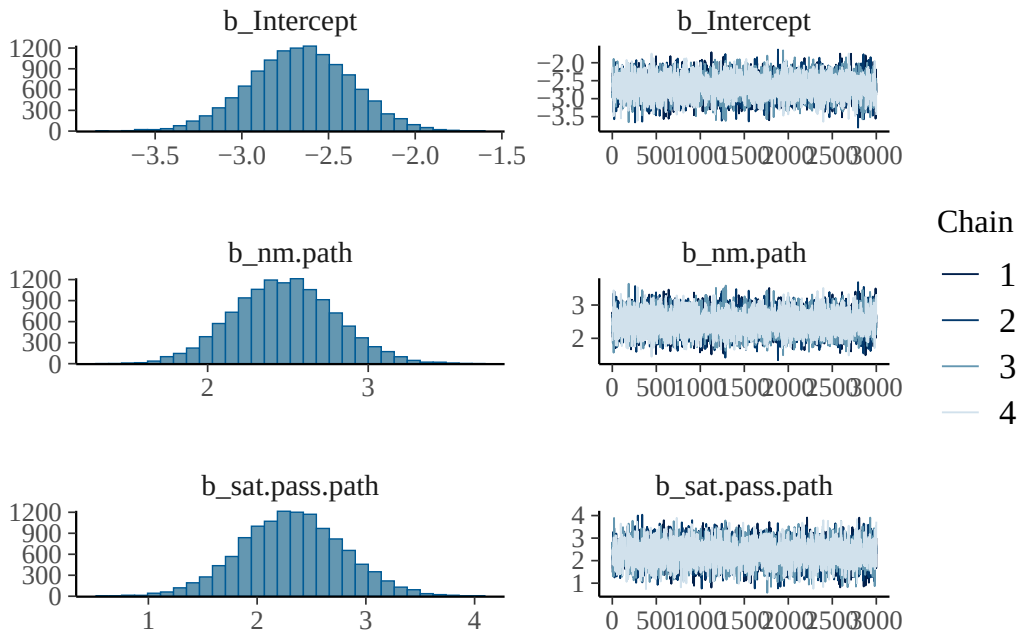
```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```




```r
#Gives us the summary of the model
print(path.fem.spawning, digits = 4)
```

```
 Family: binomial
  Links: mu = logit
Formula: total_female_spawning | trials(total_female_visits) ~ 1 + nm.path + sat.pass.path
         autocor ~ tructure(list(), class = "formula", .Environment = <environment>)
   Data: path.visits.norm.rmv (Number of observations: 137)
  Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup draws = 12000

Regression Coefficients:
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
Intercept      -2.6640    0.2917  -3.2472  -2.1037 1.0002     5578     6280
nm.path         2.4779    0.3166   1.8653   3.1101 1.0002     6020     6871
sat.pass.path   2.3227    0.4761   1.3770   3.2646 1.0002     6549     7252

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
#Plots the Markov chains
plot(path.fem.spawning)
```



Let's create a model that will inform us about each pathway.

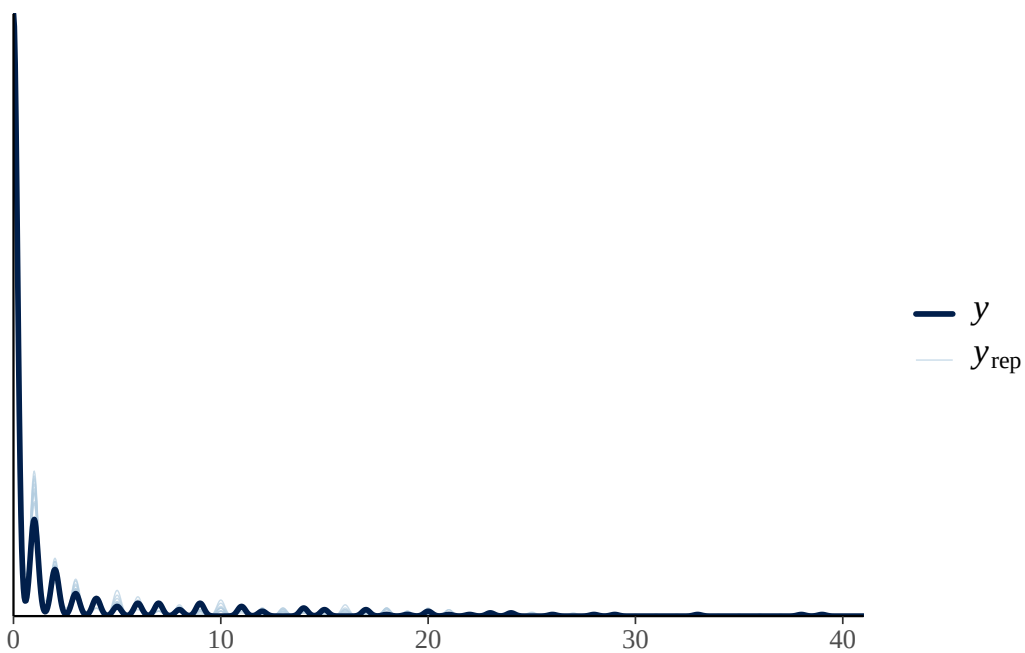
First we create the pathways

```
pathway.spawns.modl <- behavior_sheet_cln |>
  #Remove nests where there are data inconsistencies
 filter_out((female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sat_brought_by
  filter_out((female_spawning_sat_brought + females_spawing_sat_brought_sat_only_spawn) > (to
  filter_out((female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to_nm_but_only_sa
  filter_out((female_spawning_snk_brought + female_spawning_snk_brought_skn_only_spawn) > (to
   #Create new variables that account for the pathway's proportion
  mutate(
    'Nesting Male' = (total_spawns_nm_brought + total_spawns_nm_brought_sat_only_spawn),
    'Satellite Male' = (total_spawns_sat_brought + total_spawns_sat_brought_sat_only_spawn),
    'Satellite Pass' = (total_spawns_sat_passed_to_nm + total_spawns_sat_passed_to_nm_but_sat
    'Sneaker Male' = (total_spawns_snk_brought + total_spawns_snk_brought_snk_only_spawn)) |>
  #Move the pathways columns we just created into their own column for inputting into the mod
  pivot_longer(cols = c(`Nesting Male`:`Sneaker Male`),
               names_to = 'Pathway',
               values_to = 'number.spawned')
```

Then we create the model that uses total spawns at a nest as the number of trials

```
pathway.spawns.mdl.1 <- pathway.spawns.modl |>
  brm(
    family = binomial(link = 'logit'), #Tell model what type of distribution
    #Specifying the model
    formula = bf(number.spawned | trials(total_spawns) ~ 0 + Pathway + (1|nest_name)),
    prior = c(prior(normal(0, 2), class = b)),
    iter = 4000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Data Analysis Outputs/pathway.spawns.mdl.1') #saves it as an output

#Gives a plot of the model's predictions
pp_check(pathway.spawns.mdl.1)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

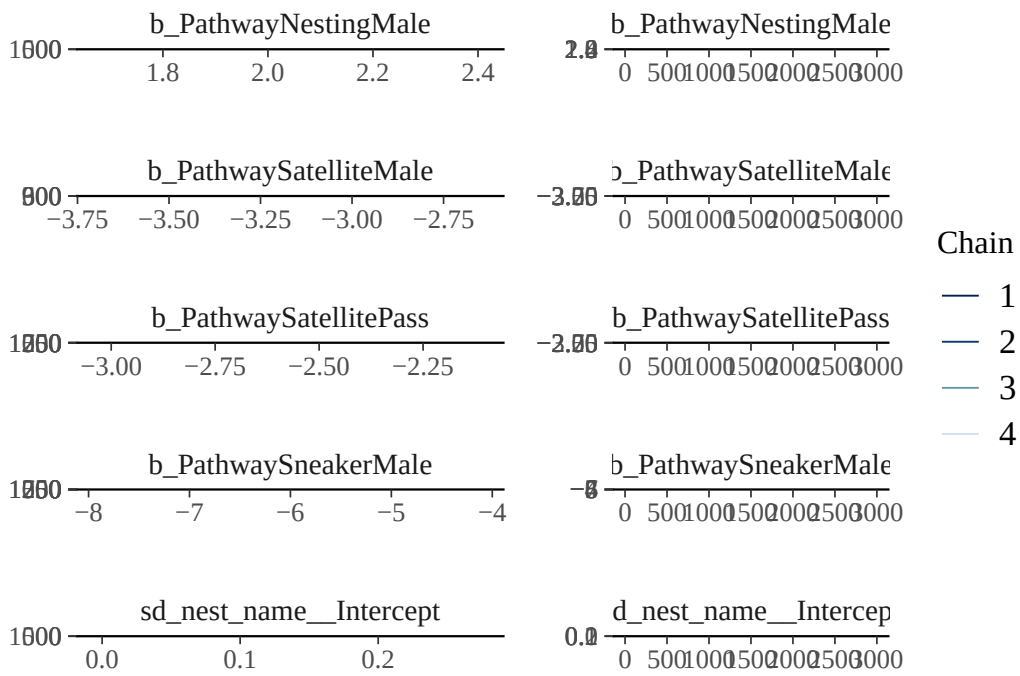



```
#Gives us the summary of the model
print(pathway.spawns.mdl.1, digits = 4)
```

```
 Family: binomial
```

```
  Links: mu = logit
Formula: number.spawned | trials(total_spawns) ~ 0 + Pathway + (1 | nest_name)
   Data: pathway.visits.modl (Number of observations: 560)
  Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup draws = 12000

Multilevel Hyperparameters:
~nest_name (Number of levels: 115)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.0514    0.0399   0.0021   0.1478 1.0000     9000     5985

Regression Coefficients:
                     Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS
PathwayNestingMale     2.0361    0.0922   1.8570   2.2198 1.0017    24982
PathwaySatelliteMale  -3.1463    0.1496  -3.4514  -2.8637 1.0000    28505
PathwaySatellitePass  -2.5567    0.1140  -2.7844  -2.3404 1.0004    26559
PathwaySneakerMale    -5.6910    0.4925  -6.7405  -4.8256 1.0003    23380
                     Tail_ESS
PathwayNestingMale       8183
PathwaySatelliteMale     8053
PathwaySatellitePass     7564
PathwaySneakerMale       7997

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#Plots the Markov chains
plot(pathway.spawns.mdl.1)
```



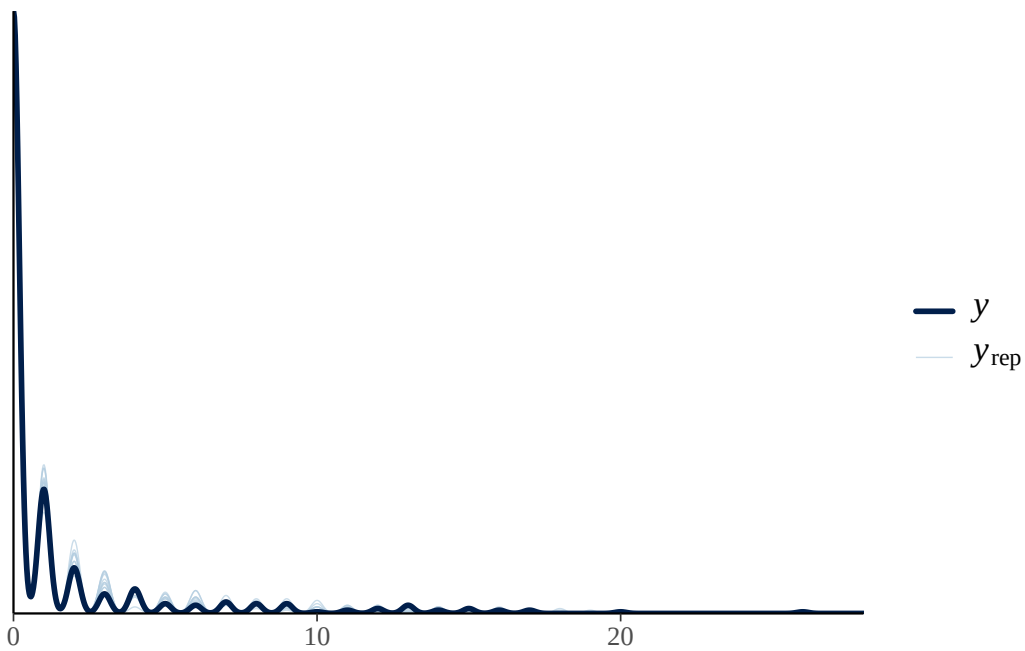
Now with visits

```
pathway.visits.modl <- behavior_sheet_cln |>
  #Remove nests where there are data inconsistencies
  filter_out((female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sat_brought_b
  filter_out((female_spawning_sat_brought + females_spawing_sat_brought_sat_only_spawn)/(tota
  filter_out((female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to_nm_but_only_sa
  filter_out((female_spawning_snk_brought + female_spawning_snk_brought_skn_only_spawn)/(tota
  #Create new variables that account for the pathway's proportion
  mutate(
    'Nesting Male' = (female_spawning_with_nm_brought_by_nm + female_spawning_with_only_sat_b
    'Satellite Male' = (female_spawning_sat_brought + females_spawing_sat_brought_sat_only_sp
    'Satellite Male Pass' = (female_spawning_sat_passed_to_nm + female_spawning_sat_passed_to
    'Sneaker Male' = (female_spawning_snk_brought + female_spawning_snk_brought_skn_only_spaw
  ) |>
  #Move the pathways columns we just created into their own column for inputting into the mod
  pivot_longer(cols = c(`Nesting Male`:`Sneaker Male`),
               names_to = 'Pathway',
               values_to = 'number.spawned')
```

Model

```r
pathway.spawns.mdl.nest <- pathway.visits.modl |>
  brm(
    family = binomial(link = 'logit'), #Tell model what type of distribution
    #Specifying the model
    formula = bf(number.spawned | trials(total_female_visits) ~ 0 + Pathway + (1|nest_name))
    prior = c(prior(normal(0, 2), class = b)),
    iter = 4000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
    sample_prior = 'yes', #Allows us to run hypothesis tests of the different Pathways
      file = 'Data Analysis Outputs/pathway.spawns.mdl.nest') #saves it as an output

#Gives a plot of the model's predictions
pp_check(pathway.spawns.mdl.nest)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

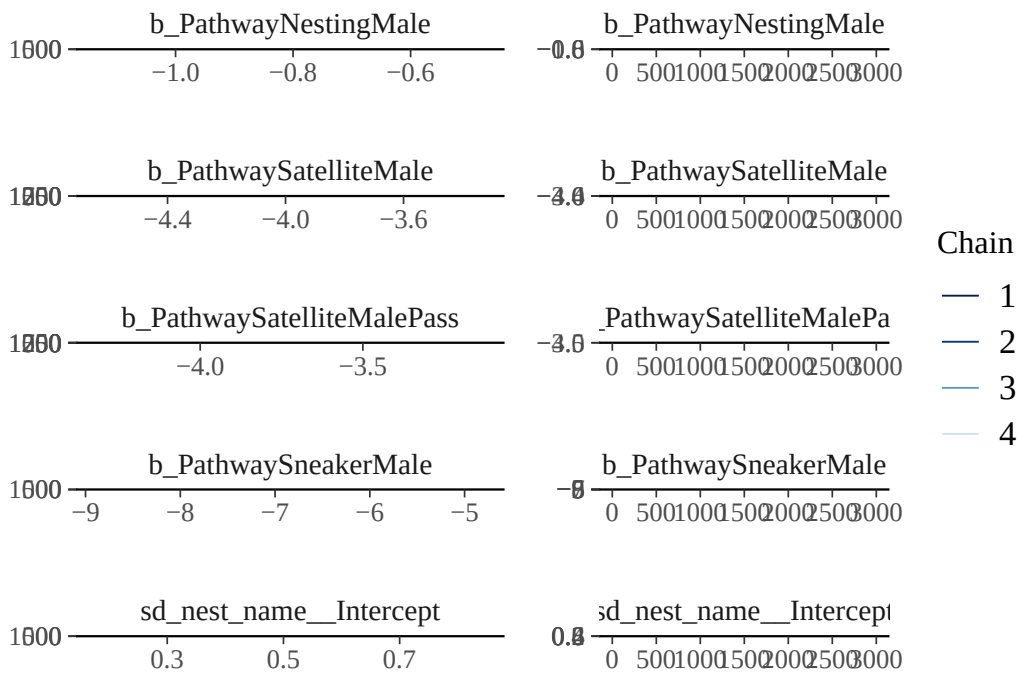



```r
#Gives us the summary of the model
print(pathway.spawns.mdl.nest, digits = 4)
```

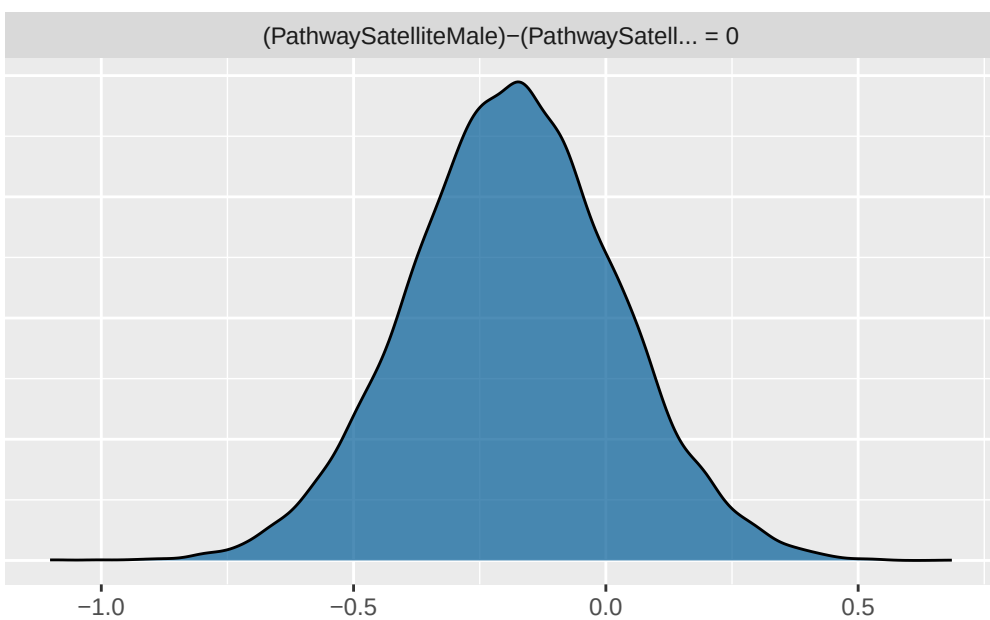
```
 Family: binomial
  Links: mu = logit
```

```
Formula: number.spawned | trials(total_female_visits) ~ 0 + Pathway + (1 | nest_name)
   Data: pathway.visits.modl (Number of observations: 552)
  Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup draws = 12000

Multilevel Hyperparameters:
~nest_name (Number of levels: 113)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.5033    0.0764   0.3633   0.6614 1.0002     4576     7714

Regression Coefficients:
                         Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS
PathwayNestingMale        -0.7551    0.0739  -0.9013  -0.6128 1.0004    10695
PathwaySatelliteMale      -3.8650    0.1613  -4.1906  -3.5607 1.0002    17842
PathwaySatelliteMalePass  -3.6807    0.1503  -3.9848  -3.3964 1.0004    18246
PathwaySneakerMale        -6.2933    0.4822  -7.3479  -5.4461 1.0001    23032
                         Tail_ESS
PathwayNestingMale          10543
PathwaySatelliteMale         9384
PathwaySatelliteMalePass     9608
PathwaySneakerMale           6775

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#Plots the Markov chains
plot(pathway.spawns.mdl.nest)
```




```
#This is how we are going to check if the Satellite pathways are statistically significantly
sat.vs.satpath <- hypothesis(pathway.spawns.mdl.nest, 'PathwaySatellite Male = PathwaySatelli
plot(sat.vs.satpath)
```



This hypothesis test tells us that the two satellite male pathways are not statistically significantly different from one another.

Now lets back-transform our data

```
back.transformed <- data.frame('Estimate' = c(plogis(-0.7551),plogis(-3.8650), plogis(-3.680

back.transformed.summary <- back.transformed |>
  mutate('l-95% CI' = c(plogis(-0.9013), plogis(-4.1906), plogis(-3.9848), plogis(-7.3479)))
  mutate('u-95% CI' = c(plogis(-0.6128), plogis(-3.5607), plogis(-3.3964), plogis(-5.4461)))
back.transformed.summary
```

```
                      Estimate     l-95% CI   u-95% CI
Nesting Male        0.319711053 0.2887834203 0.35142074
Satellite Male      0.020532501 0.0149114817 0.02763361
Satellite Male Pass 0.024585636 0.0182566594 0.03240816
Sneaker Male        0.001845238 0.0006435288 0.00429457
```

So based on our models it seems like there is an effect on spawning for each of the pathways (albeit practically non-existent in some). Given our data it seems that for:

The probability of a female spawning when brought by the nesting male is around 31.9%

The probability of a female spawning when brought by the satellite male is around 2.05%

The probability of a female spawning when brought by the satellite male then passed to the nesting male is around 2.46%

The probability of a female spawning when brought by the sneaker male is 0.1%.

Since none of our 95% confidence intervals overlap with zero, we can assume that these effects are different from zero (again based on our data).

Thinking about this, it makes sense. As we saw in the graphs, the majority of female visits are by the nesting male, thus it would have a higher probability than the other pathways. It is interesting that the satellite male pathways are so low, however, the pathway when the satellite male passes the female to the nesting male being higher than when it's just the satellite male is super interesting, and is going to be something we're going to continue to look at. Overall, these visualizations and models help give us a foundation for what we're interested in!

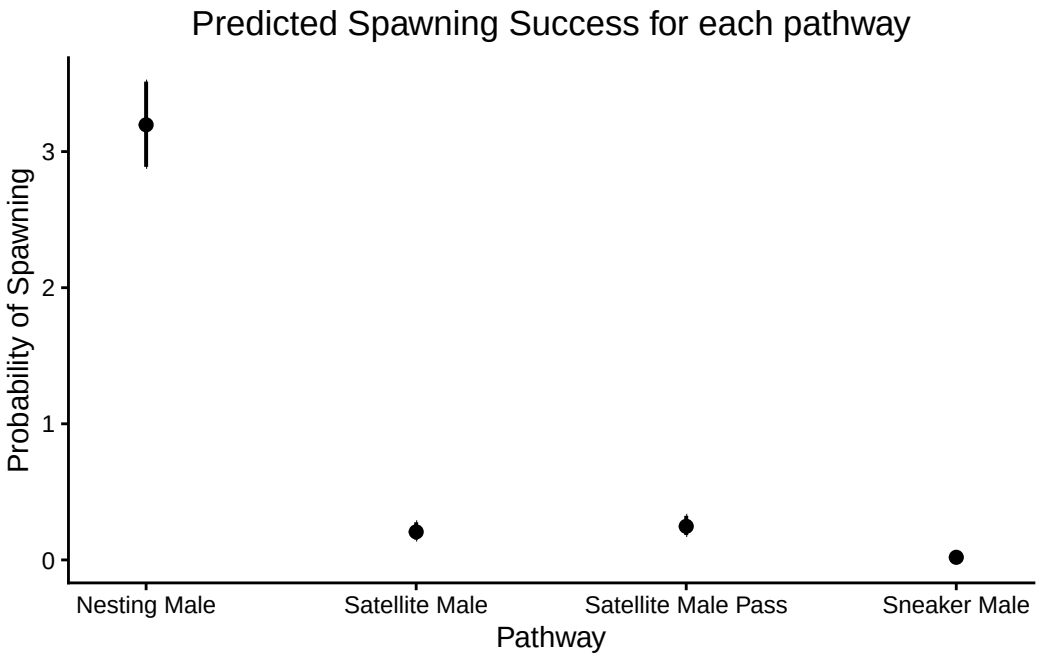
This next chunk is the graph of our model's predictions

```
#Create the predictions
preds <- predict_response(pathway.spawns.mdl.nest, terms = c('Pathway'))
```

```
You are calculating adjusted predictions on the population-level (i.e.
  `type = "fixed"`) for a *generalized* linear mixed model.
  This may produce biased estimates due to Jensen's inequality. Consider
  setting `bias_correction = TRUE` to correct for this bias.
  See also the documentation of the `bias_correction` argument.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Pathway`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```r
#Plots the prediction
plot(ggpredict(pathway.spawns.mdl.nest, # Tells the data
               c("Pathway"), #Which term is on the x-axis
               type = "fixed", #it is fixed
               condition = c(total_female_visits = 10) #How many "trials to show"
     )) +
  theme(legend.position = "none") + #Removes any legend
  theme_classic() + # Sets the overall theme
  labs(y = 'Probability of Spawning',
       title = 'Predicted Spawning Success for each pathway') + #Adjust axis and title
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle
```

```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Pathway`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```



This tells us some interesting information. Given 10 random trials, we would expected around 3 females to spawn when brought by the Nesting Male, zero for Sneaker males, and almost zero for the other types, this prediction makes sense. Since these pathways are counting when the Nesting Male and Satellite Male are openly spawning, but not all the sneak spawns from sneaker and satellites.

Final notes

You are now caught up with our where our research stands! While our findings thus far can seem "disappointing" or "lack-luster" that is actually good! Nature is so complex, let alone animal behavior, thus any findings provide useful information! For our research, these models indicate that we need to adjust how we collect our behavior data. So far, we've only been working on total spawns of each pathway and not looking at how/what influences individual females (because this is going to be a lot of work, that's where you come in!). Additionally, while it is small, and not statistically significant, the difference between the satellite male and the satellite male pass pathway is promising! Maybe, with some adjustment to how we are quantifying our data, we can test if the different pathways matter. Which would be very interesting because that means the females are paying attention to the social context around them!

Session info

Here is the session info!

```r
sess <- sessionInfo()
sess
```

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sonoma 14.6.1

Matrix products: default
BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framewo
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] MASS_7.3-65        RColorBrewer_1.1-3 brms_2.23.0        Rcpp_1.1.1
 [5] readxl_1.4.5       here_1.0.2         janitor_2.2.1      lubridate_1.9.4
 [9] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.0        purrr_1.2.1
[13] readr_2.1.6        tidyr_1.3.2        tibble_3.3.1       tidyverse_2.0.0
[17] ggeffects_2.3.2    ProbBayes_1.1      shiny_1.12.1       gridExtra_2.3
[21] ggplot2_4.0.2      LearnBayes_2.15.2

loaded via a namespace (and not attached):
 [1] Rdpack_2.6.6          inline_0.3.21         sandwich_3.1-1
 [4] rlang_1.1.7           magrittr_2.0.4        snakecase_0.11.1
 [7] otel_0.2.0            matrixStats_1.5.0     compiler_4.5.2
[10] mgcv_1.9-3            loo_2.9.0             reshape2_1.4.5
[13] callr_3.7.6           vctrs_0.7.1           pkgconfig_2.0.3
[16] fastmap_1.2.0         backports_1.5.0       labeling_0.4.3
[19] utf8_1.2.6            promises_1.5.0        rmarkdown_2.30
[22] tzdb_0.5.0            nloptr_2.2.1          haven_2.5.5
[25] ps_1.9.1              xfun_0.56             jsonlite_2.0.0
```

```
[28] later_1.4.5           parallel_4.5.2        R6_2.6.1
[31] stringi_1.8.7         StanHeaders_2.32.10   boot_1.3-32
[34] numDeriv_2016.8-1.1   cellranger_1.1.0      rstan_2.32.7
[37] knitr_1.51            zoo_1.8-15            bayesplot_1.15.0
[40] glmmTMB_1.1.14        httpuv_1.6.16         Matrix_1.7-4
[43] splines_4.5.2         timechange_0.4.0      tidyselect_1.2.1
[46] rstudioapi_0.18.0     abind_1.4-8           yaml_2.3.12
[49] TMB_1.9.19            codetools_0.2-20      curl_7.0.0
[52] processx_3.8.6        pkgbuild_1.4.8        plyr_1.8.9
[55] lattice_0.22-7        bayestestR_0.17.0     withr_3.0.2
[58] bridgesampling_1.2-1  S7_0.2.1              posterior_1.6.1
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[64] RcppParallel_5.1.11-1 pillar_1.11.1         tensorA_0.36.2.1
[67] checkmate_2.3.4       stats4_4.5.2          reformulas_0.4.4
[70] insight_1.4.6         distributional_0.6.0  generics_0.1.4
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[76] scales_1.4.0          minqa_1.2.8           xtable_1.8-4
[79] glue_1.8.0            tools_4.5.2           lme4_1.1-38
[82] mvtnorm_1.3-3         grid_4.5.2            rbibutils_2.4.1
[85] QuickJSR_1.9.0        datawizard_1.3.0      colorspace_2.1-2
[88] nlme_3.1-168          cli_3.6.5             Brobdingnag_1.2-9
[91] V8_8.0.1              gtable_0.3.6          digest_0.6.39
[94] farver_2.1.2          htmltools_0.5.9       lifecycle_1.0.5
[97] mime_0.13
```